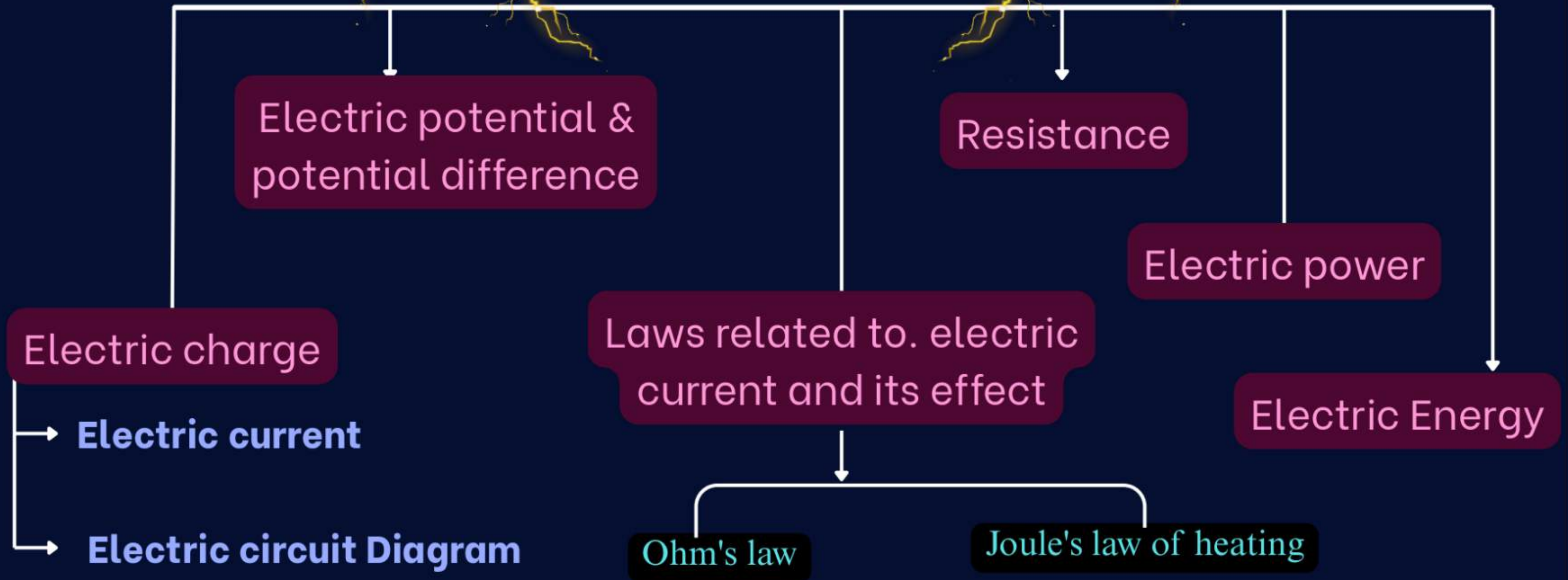


CLASS X - SCIENCE

ELECTRICITY



ELECTRICITY



ELECTRIC CHARGE

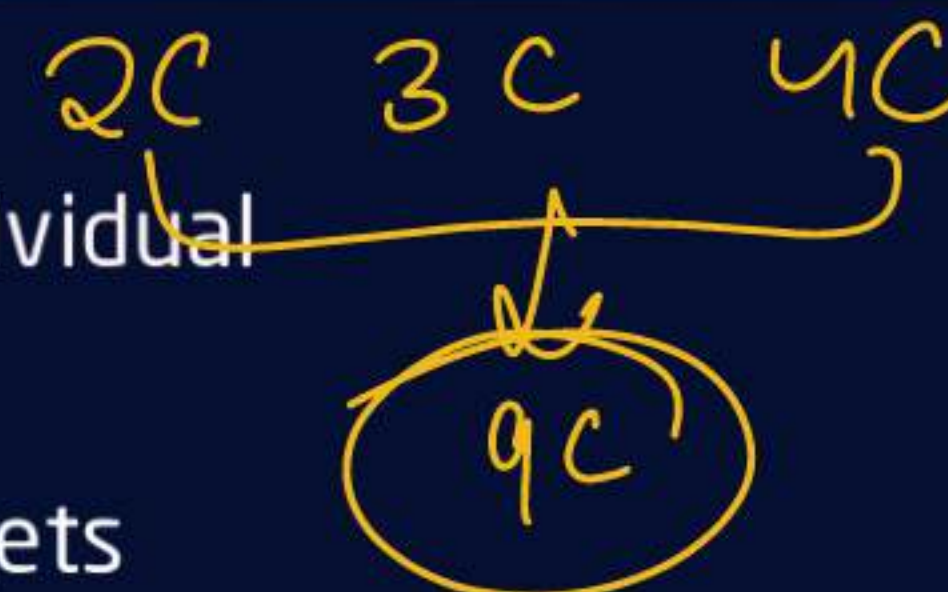
$$e^{\ominus} =$$

- Electric charge is a fundamental property of matter due to which matter particles can be positive or negative in nature
- There are two types of charges: positive and negative.  
- Like charges repel each other and unlike charges attract each other.



- *The SI unit of charge is coulomb (C).*

1. Additivity of Charges: The total charge on a system is the algebraic sum of individual charges.



2. Conservation of Charge: Charge can neither be created nor destroyed, it only gets transferred.

3. Quantisation of Charge:

$$e = -1.6 \times 10^{-19} \text{ C}$$

- Any observable charge on a body is always an integral multiple of the elementary charge e .

- This can be written as:

$$Q = ne$$



$$Q = \pm ne$$

q = total charge,
 n = integer (1, 2, 3 ...),
 $e = 1.6 \times 10^{-19} \text{ C}$

(2014, 2017, 2023 Sample)

Q. A body has a charge of $4.8 \times 10^{-19} \text{ C}$. Calculate the number of electrons it has gained or lost.

$$Q = 4.8 \times 10^{-19}$$

$$Q = ne$$

$$4.8 \times 10^{-19} = n \times 1.6 \times 10^{-19}$$

$$n = \frac{4.8}{1.6} = 3$$

ELECTRIC CURRENT



$$I = \frac{Q}{t} \quad C/SPC$$

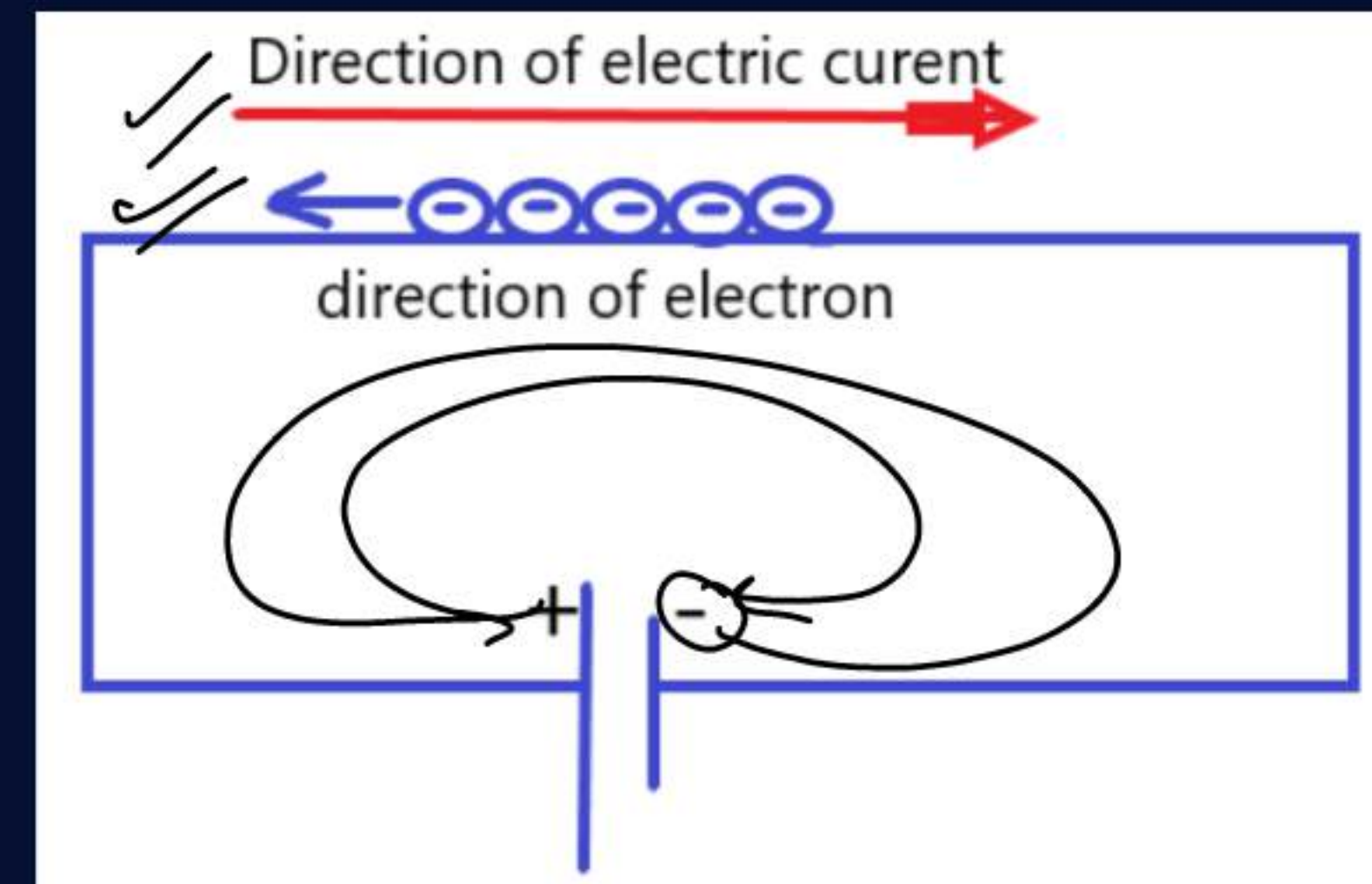
Electric current is defined as the rate of flow of charge through a cross-section of a conductor per unit time.

It is denoted by I .

SI unit: Ampere (A) or coulomb per second

$$I = \frac{Q}{T}$$

I = current,
 Q = charge,
 t = time.



#One ampere is the current flowing through a conductor when 1 coulomb of charge flows per second.

$$1 A = \frac{1 C}{1 sec}$$

Direction:

- **Conventional current** → taken from positive (+) to negative (-) terminal.
- **Actual flow** → electrons move from (-) to (+).

Smaller units:

$$1 mA = 10^{-3} A$$

$$1 \mu A = 10^{-6} A$$

$$10^{-6}$$



2017, 2021 Board Exams, Sample Paper 2023

Q. One coulomb charge flows through a conductor in 1 second. Find the current.

The handwritten solution shows the following steps:

- Given: $Q = 1\text{C}$
- Given: $t = 1\text{sec}$
- Formula: $Q = I t$ (boxed)
- Derivation: $I = \frac{Q}{t}$ (circled)
- Result: 1A (circled)

Arrows indicate the flow of information: from the given values to the formula, then to the rearranged formula, and finally to the calculated result.

NCERT, 2015, CBSE Sample Paper 2020.

Q. A current of 0.5 A is drawn by a filament of an electric bulb for 10 minutes. Find the amount of electric charge that flows through the circuit

$$Q = I t$$

$$I = 0.5 \text{ A}$$

$$t = 10 \times 60 = 600 \text{ s}$$

$$Q = 600 \times 0.5$$

$$= \underline{300 \text{ C}}$$

2013, 2015, 2017, 2020

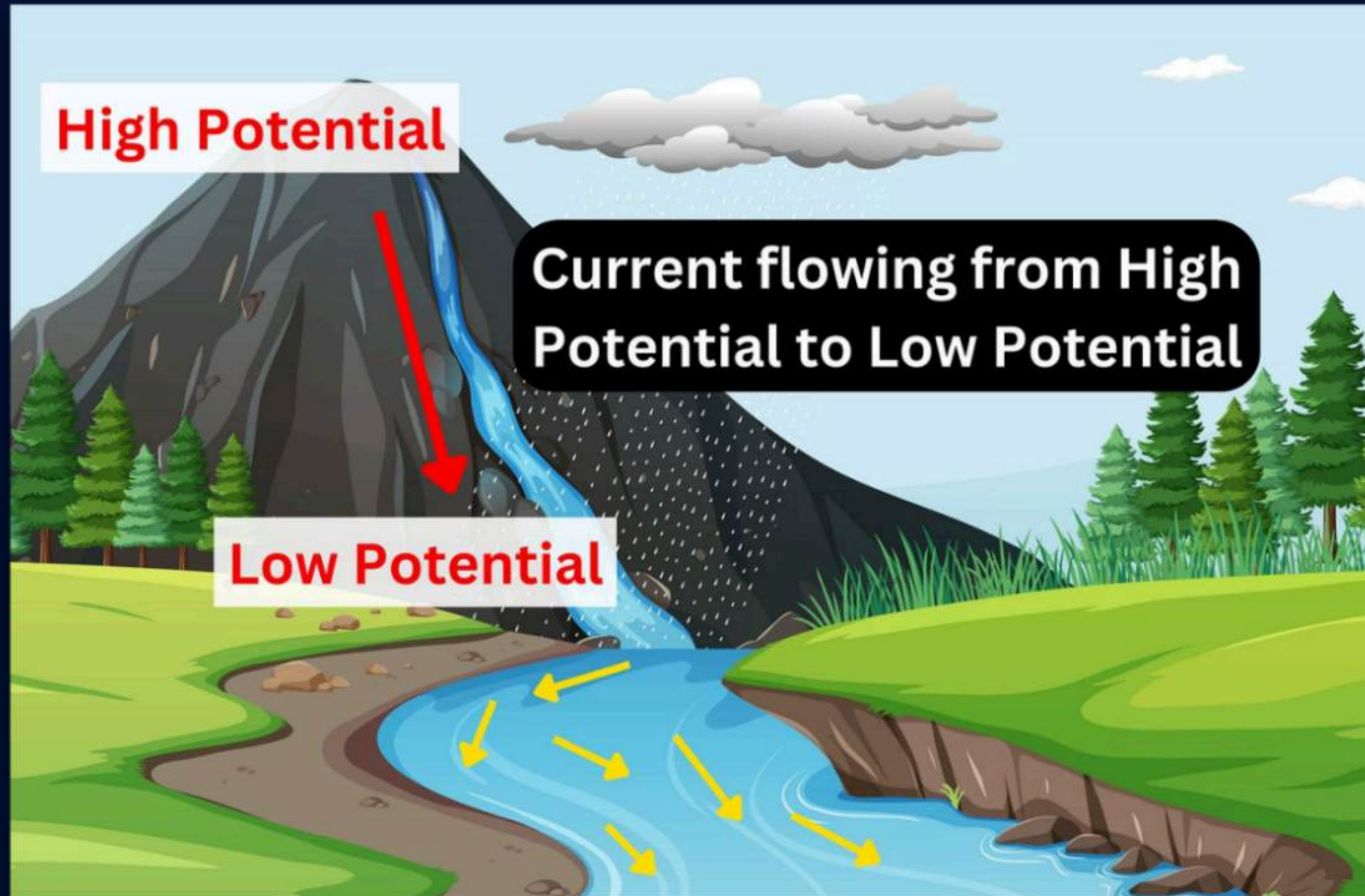
Q. A current of 10 A flows through a conductor for 2 minutes. Find the charge and number of electrons.

$$\begin{aligned} Q &= It \\ &= 10 \times 120 \\ &= \underline{1200 \text{ C}} \end{aligned}$$

$$\begin{aligned} n &= \frac{750}{\cancel{3000} \times 10^{19}} \\ &= \underline{750 \times 10^{19}} \end{aligned}$$

$$\begin{aligned} Q &= ne \\ 1200 &= n \times [1.6 \times 10^{-19}] \end{aligned}$$

POTENTIAL DIFFERENCE



Electric Potential Difference : The potential difference between two points in an electric circuit is the work done to move a unit charge from one point to another. It is the driving force that causes the flow of electric current. It is denoted by V .

SI unit: Volt (V) or joule per coulomb

$$V = \frac{W}{Q} = \text{Joule/C}$$

Volt

$$W = V \times Q$$

$$V = \frac{W}{Q}$$

$$1 \text{ volt} = \frac{1 \text{ joule}}{1 \text{ coulomb}}$$

#1 Volt: 1 Joule of work done to move 1 unit positive charge between two points.

CBSE 2019 Board Exam

Q. Calculate the amount of work done by a cell when 20 C of charge is moved through a potential difference of 3 V.

$$Q = 20 \text{ C}$$

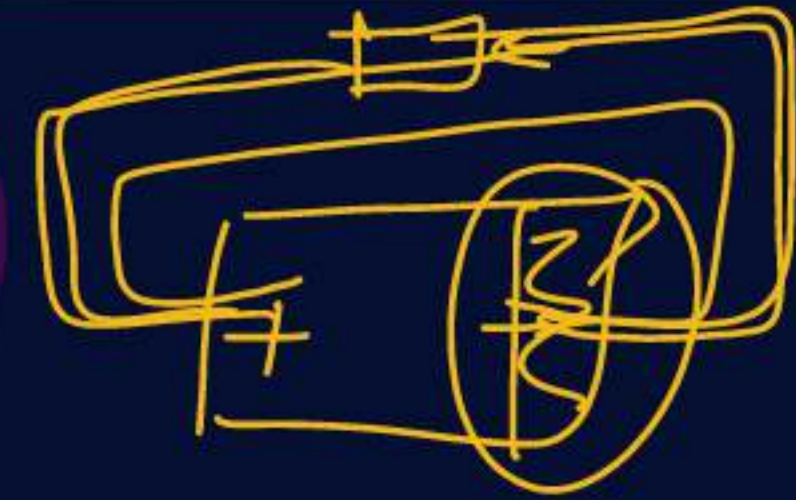
$$V = 3 \text{ V}$$

$$W$$

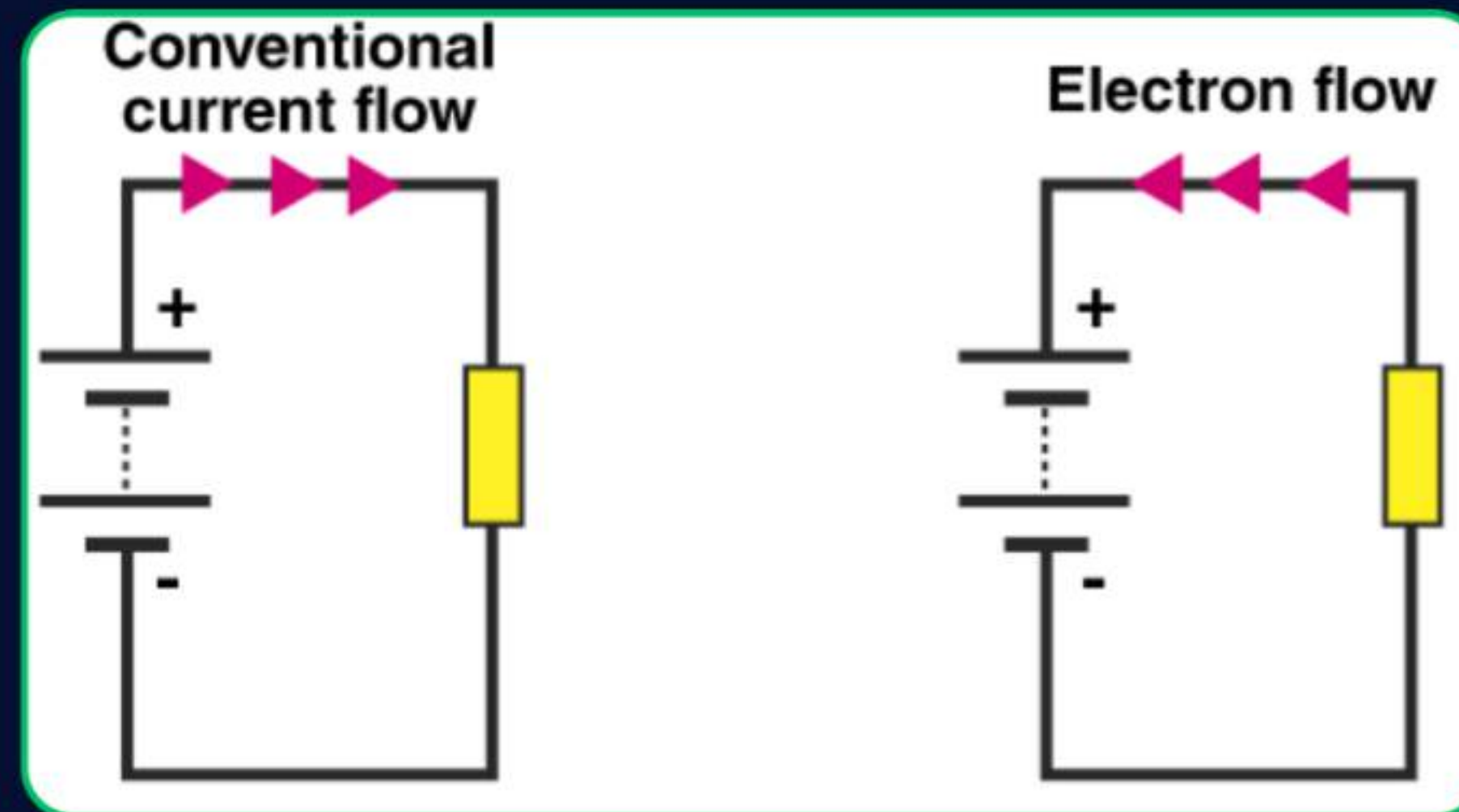
$$W = Q \times V$$

$$= 60 \text{ Joule}$$

WHY CURRENT FLOWS?



- **Electrons (e^-)** flow in a conductor from lower potential to higher potential.
- **Conventional current** flows from higher potential to lower potential.
- A cell or battery creates this potential difference (PD), which makes charges flow in a closed circuit.

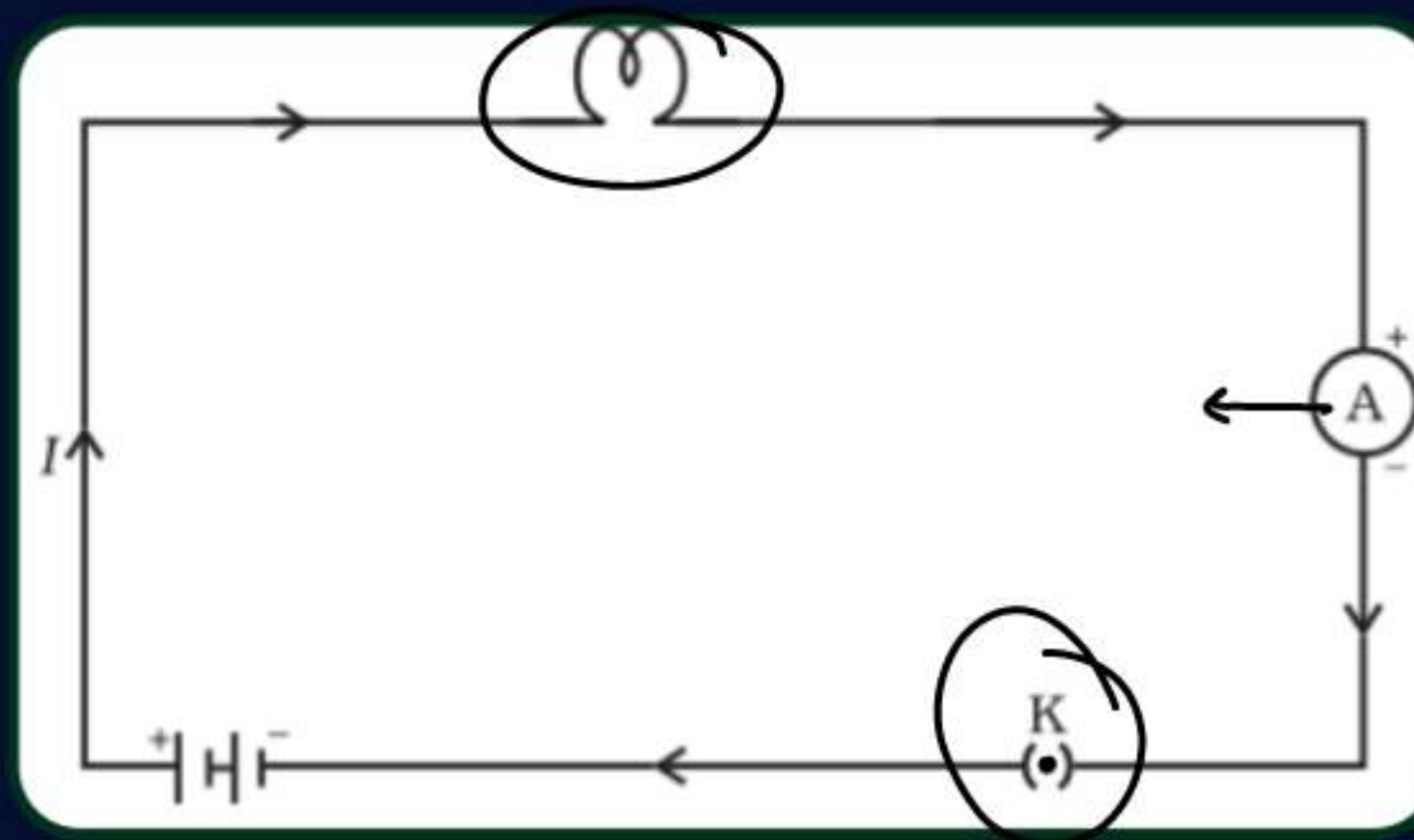


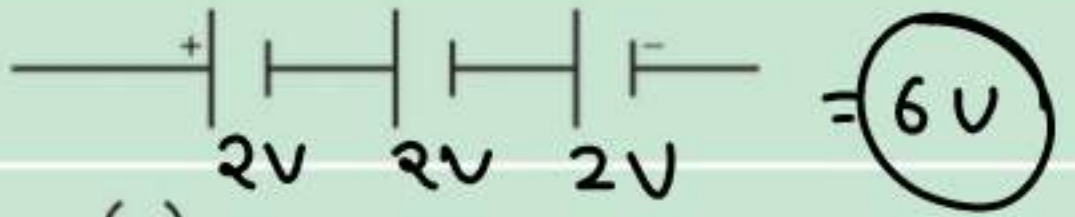
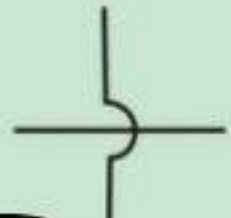
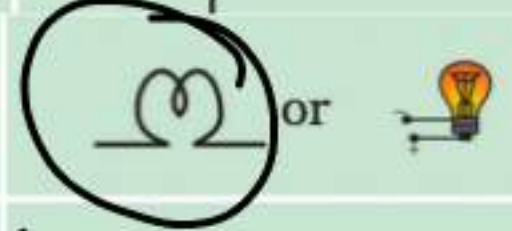
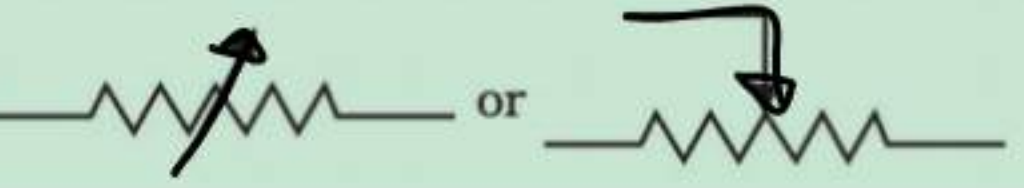
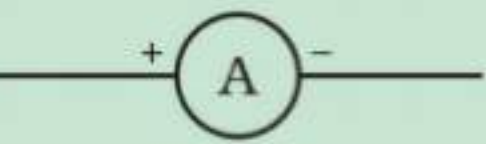
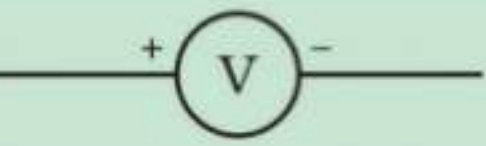
ELECTRIC CIRCUIT

A circuit diagram is a schematic drawing of an electric circuit where components are shown using standard symbols and are connected by lines (wires).

Basic Parts of an Electric Circuit

- Source: Cell/Battery (provides potential difference)
- Control: Plug key/Switch (open = OFF, closed = ON)
- Conductors: Connecting wires
- Components: e.g., resistor, bulb, ammeter, voltmeter, etc.



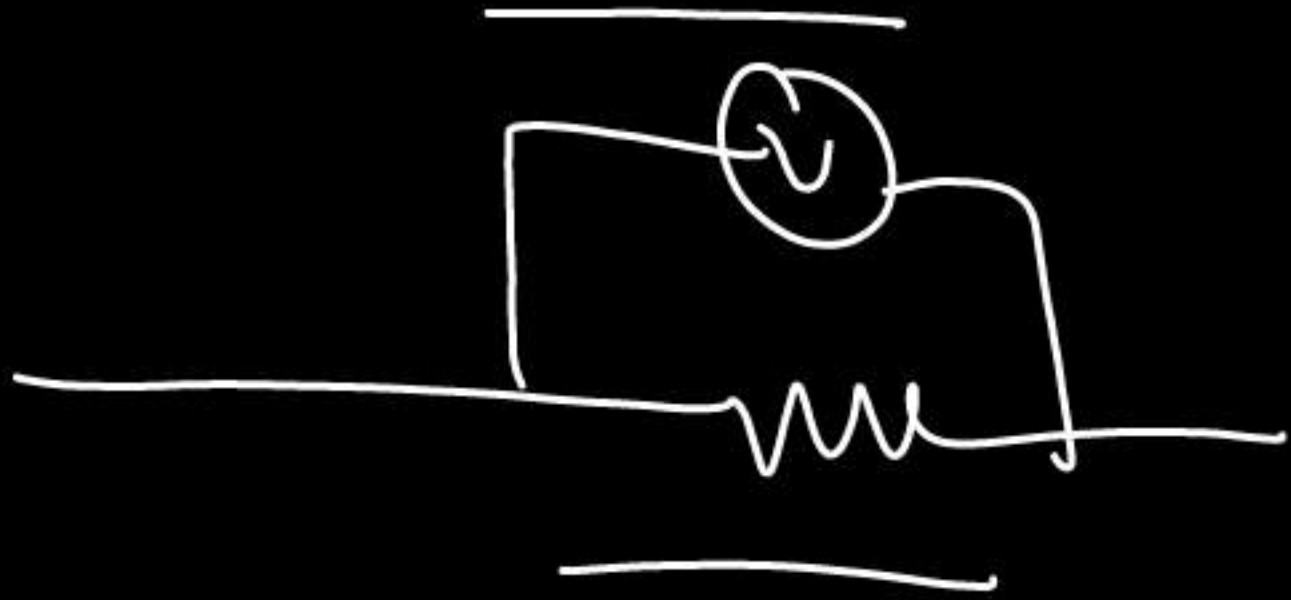
Sl. No.	Components	Symbols
1	An electric cell	
2	A battery or a combination of cells	
3	Plug key or switch (open)	
4	Plug key or switch (closed)	
5	A wire joint	
6	Wires crossing without joining	
7	Electric bulb	
8	A resistor of resistance R	
9	Variable resistance or rheostat	
10	Ammeter	
11	Voltmeter	

components in circuit diagrams

AMMETER AND VOLTMETER

Ammeter	Voltmeter
Instrument used to measure electric current in a circuit.	Instrument used to measure potential difference (voltage) between two points in a circuit.
Unit of current is ampere (A) .	Unit of potential difference is volt (V) .
Always connected in series with the circuit component.	Always connected in parallel across the circuit component.
Reason: In series, the same current flows through the ammeter and the component.	Reason: In parallel, it measures the exact potential difference across that component.
Symbol: A circle with A inside.	Symbol: A circle with V inside.





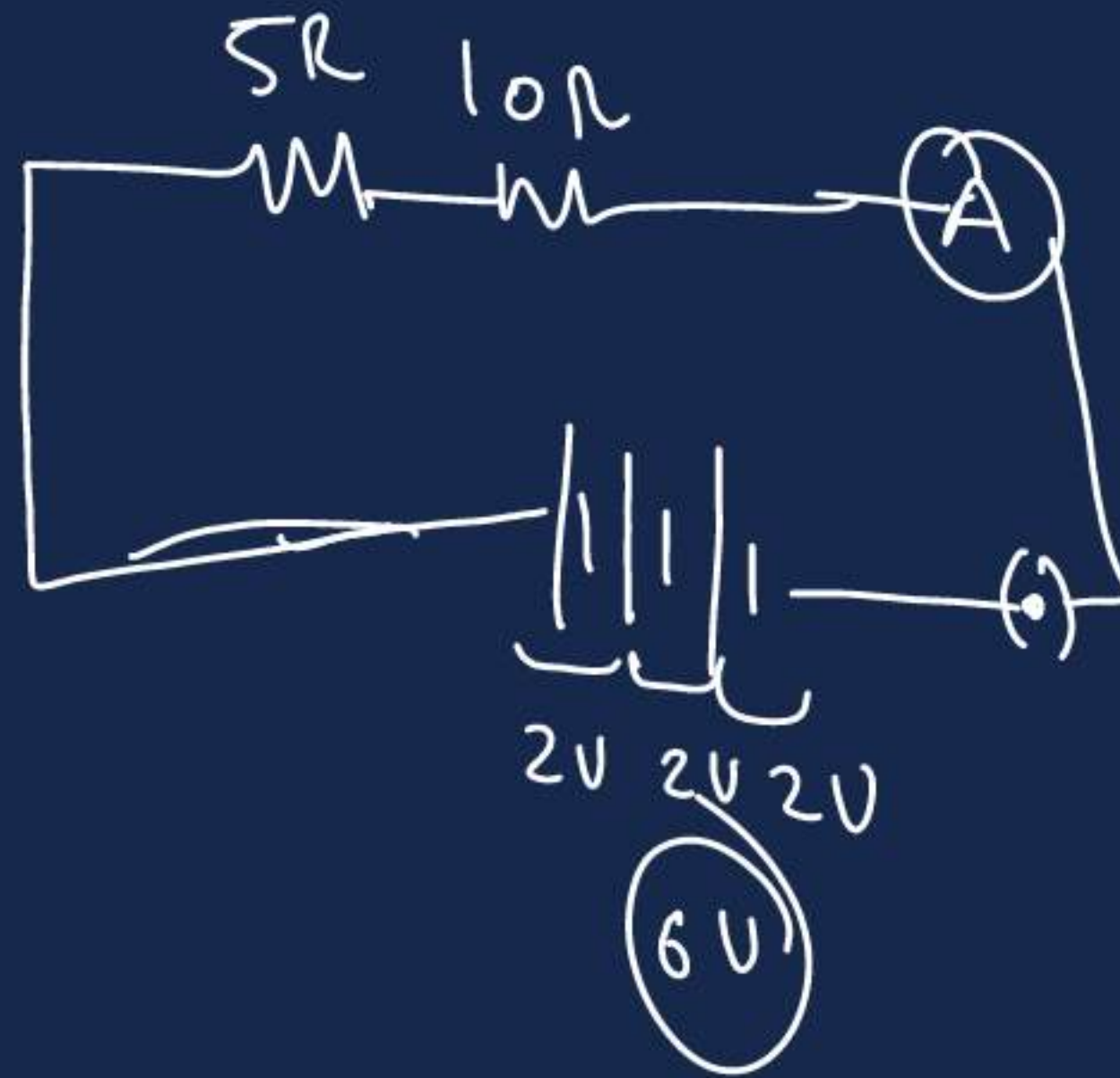
2015

Q. Identify the following symbols of commonly used components in a circuit diagram:

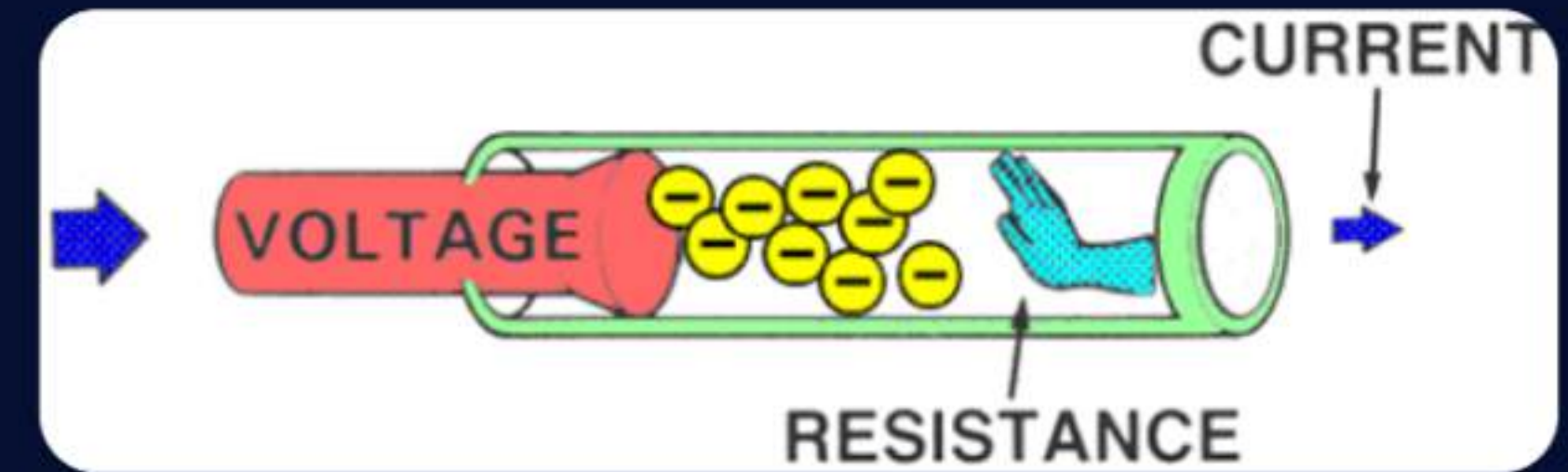


2017 Board

Q. Draw a schematic diagram of a circuit consisting of a battery of 3 cells of 2 V each, a $5\ \Omega$ resistor, a $10\ \Omega$ resistor, an ammeter, and a plug key, all connected in series.



RESISTANCE (R)



Resistance is the property of a conductor to oppose the flow of electric current through it.

$$R = \frac{V}{I}$$

- SI Unit: Ohm (Ω).

Ohm Ω

1 Ohm: Resistance of a conductor is 1Ω if a potential difference of 1 V across its ends allows a current of 1 A to flow.

A rheostat is a variable resistor used to regulate the current in an electric circuit without changing the voltage.

OHM'S LAW

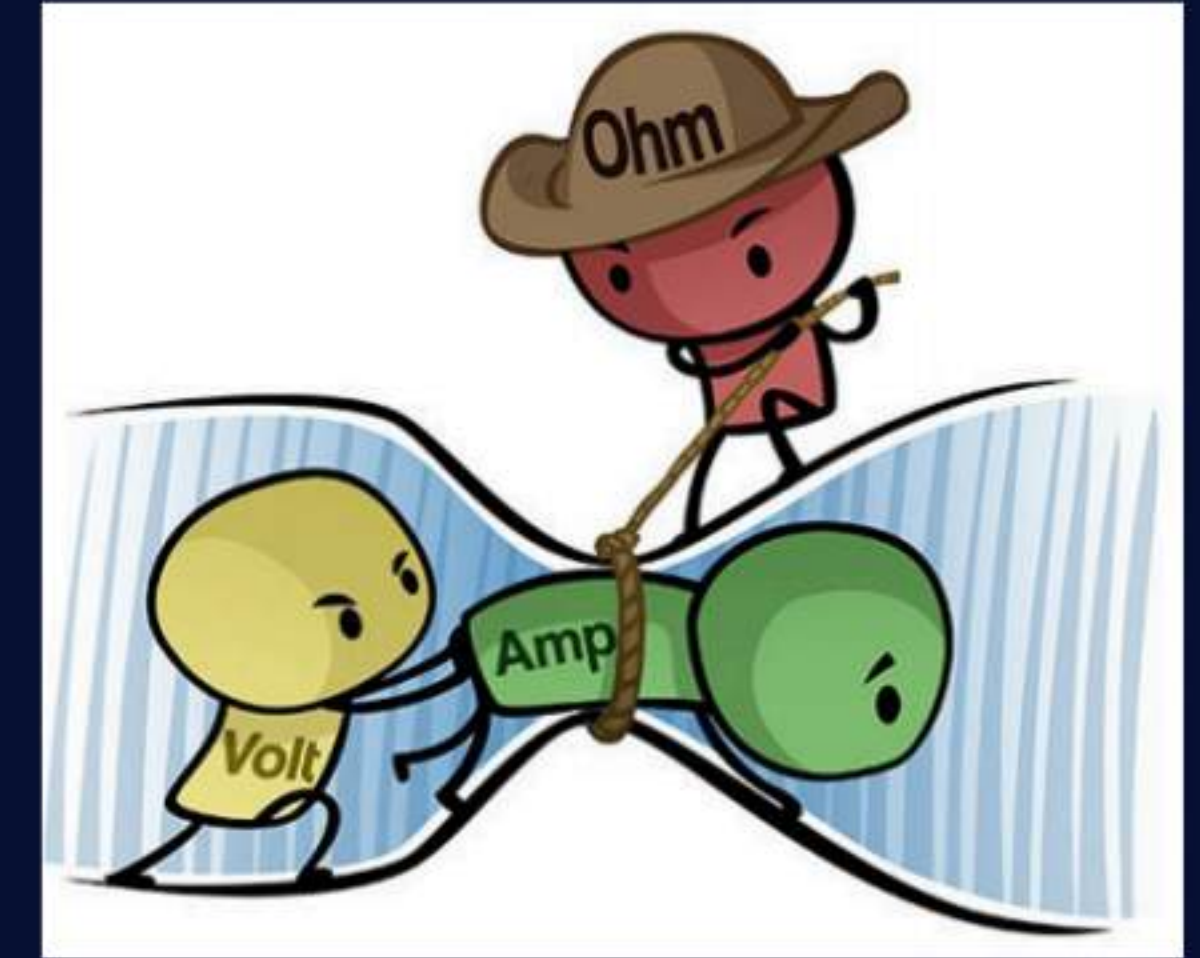


$$V \propto I$$

$$V = IR$$

“The potential difference (V) across the ends of a conductor is directly proportional to the current (I) flowing through it, provided the temperature remains constant.”

$$V \propto I$$
$$V = IR$$

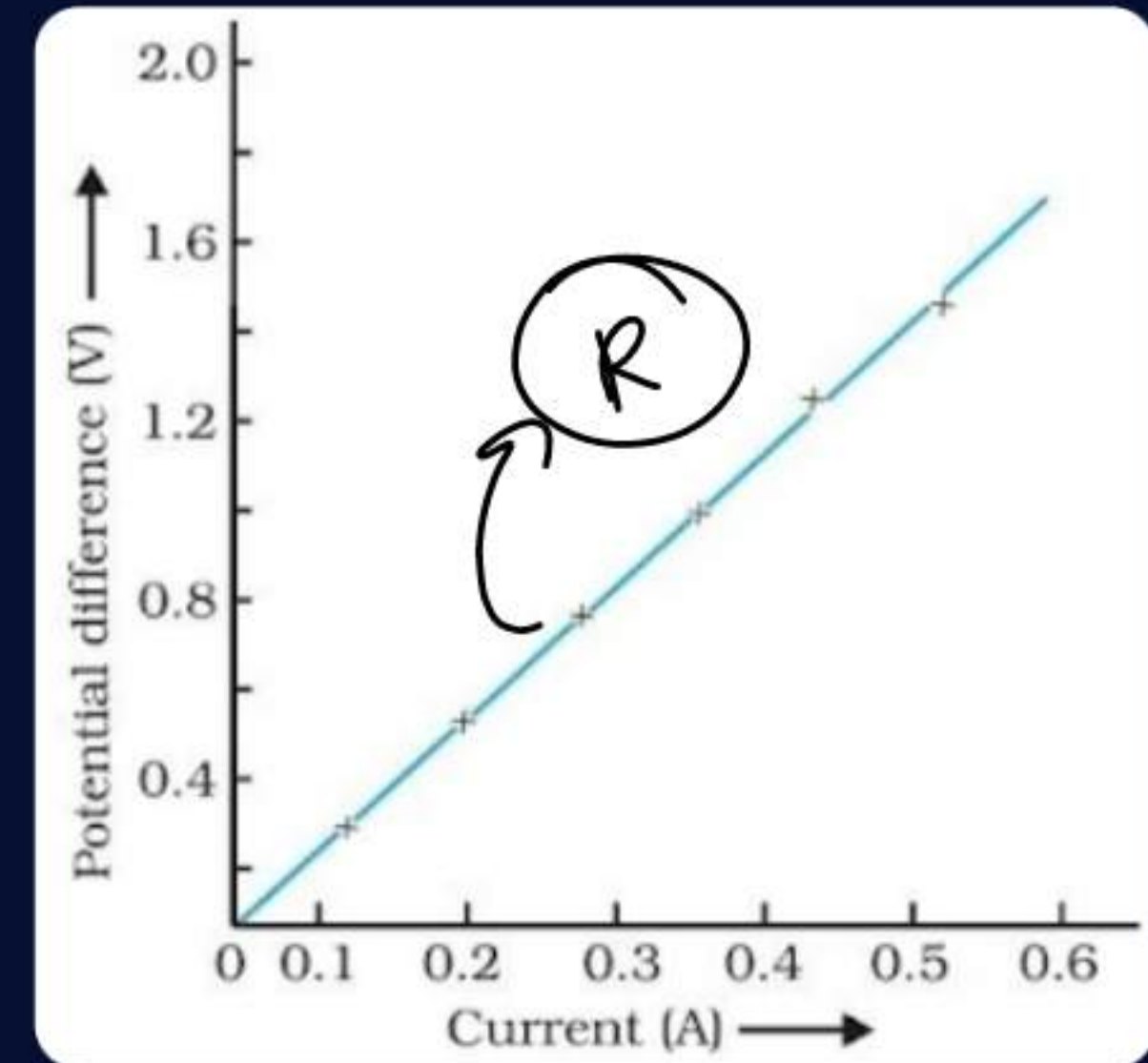


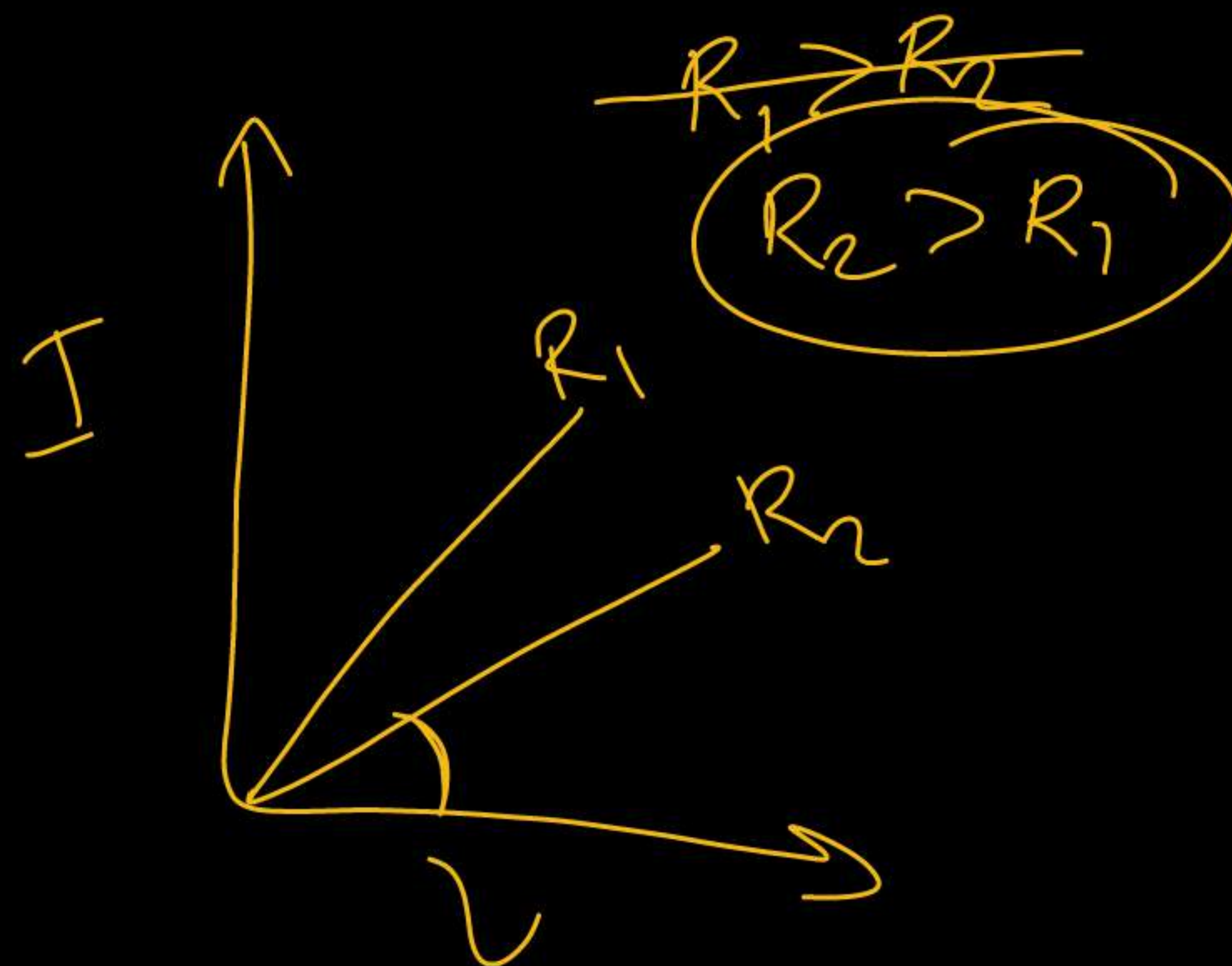
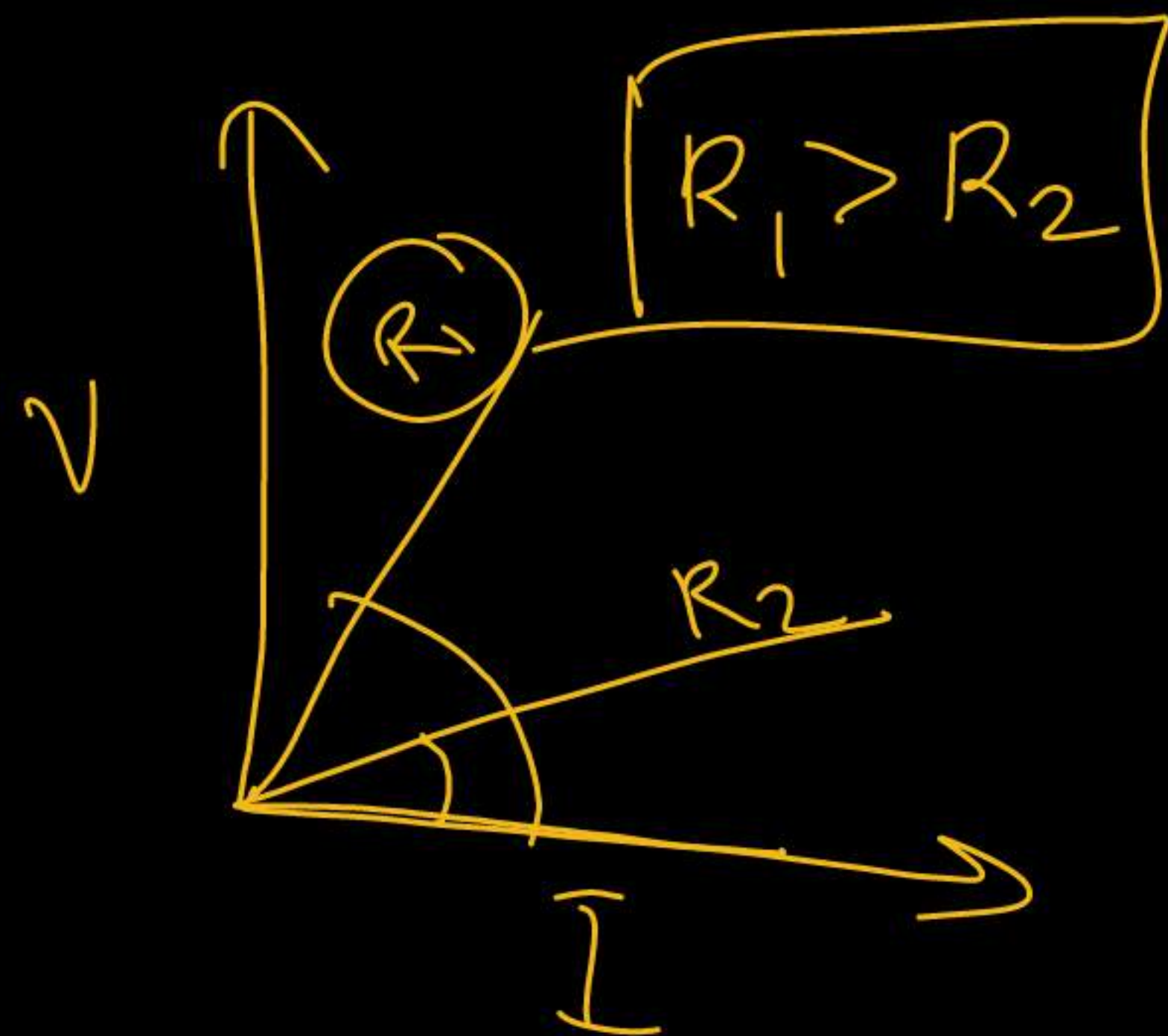
V-I CHARACTERISTIC GRAPH

When potential difference is 1 V and current through the circuit is 1 A, then resistance is 1 ohm.

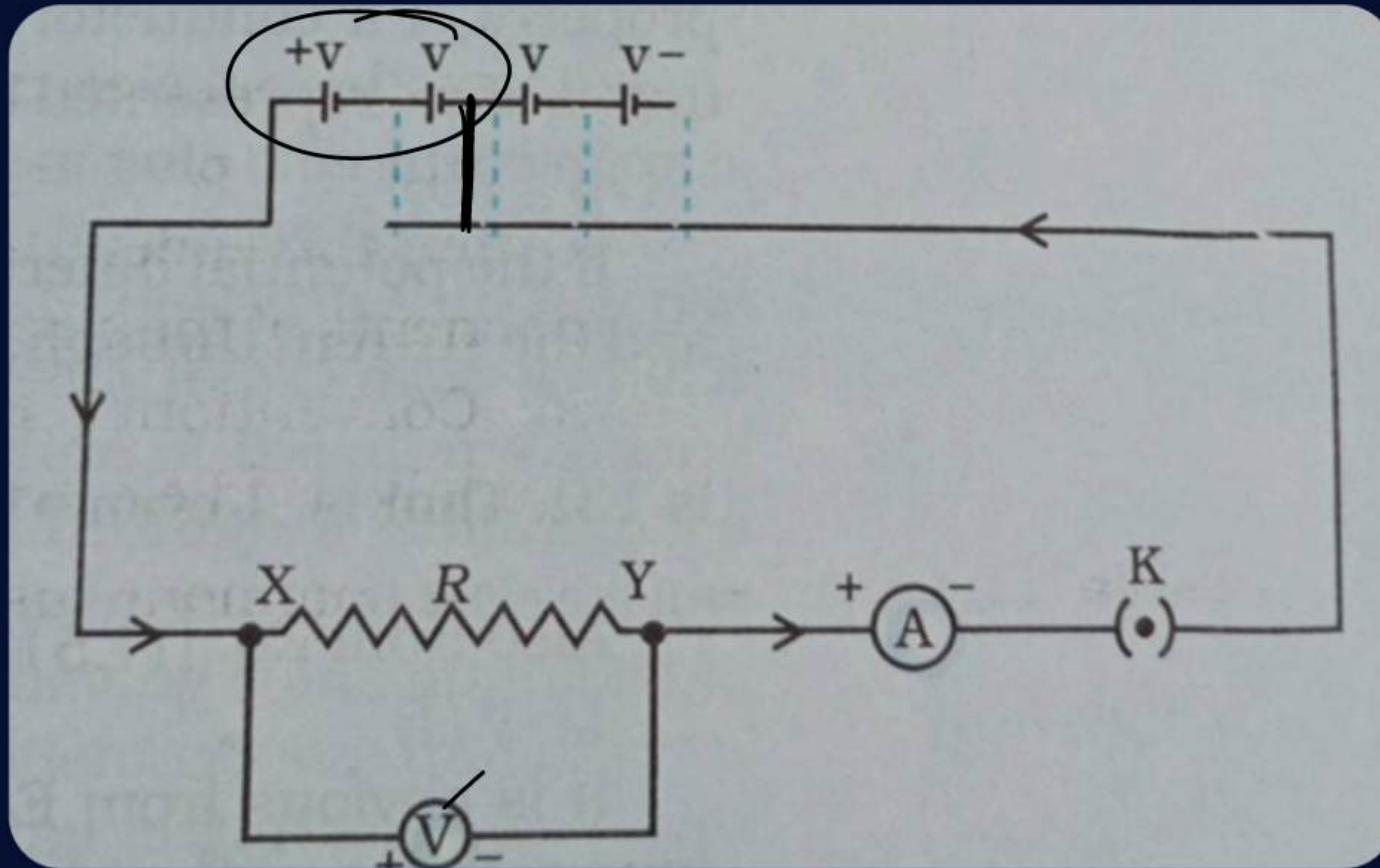
Example of V-I Characteristic Graph

- Ohmic conductor like a resistor follows Ohm's Law ($V = IR$).
- Its V-I graph is a straight line passing through the origin.
- This means voltage is directly proportional to current.
- The slope of the graph gives the resistance (R).





OHM'S LAW EXPERIMENT



V	I
2V	1A
4V	2A
6V	3A

(2019, 2023, 2024)

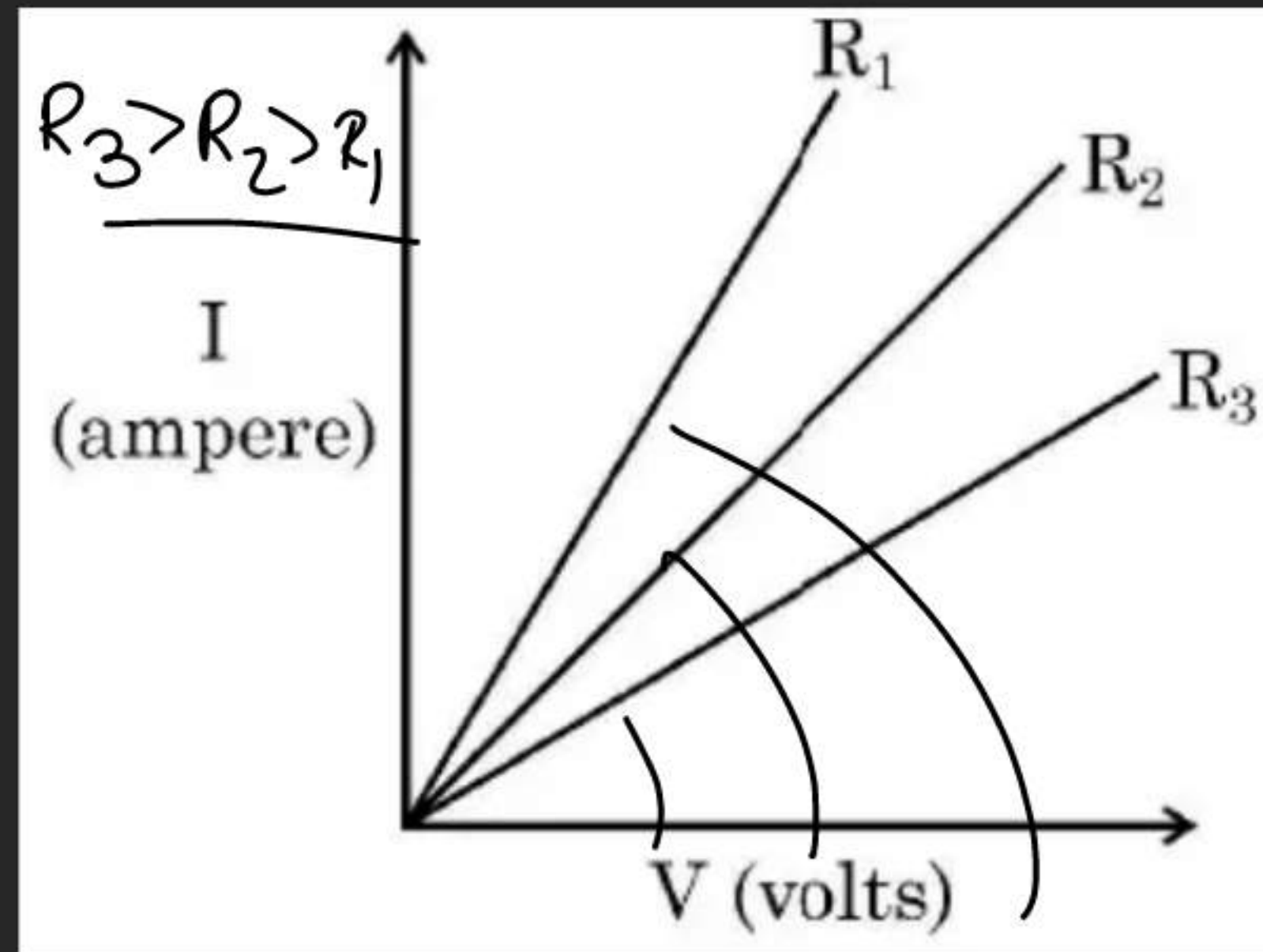
Q. Study the I-V graph for three resistors of resistances R_1 , R_2 and R_3 and select the correct statement from the following:

(a) $R_1 = R_2 = R_3$

(b) $R_1 > R_2 > R_3$

~~(c) $R_3 > R_2 > R_1$~~

(d) $R_2 > R_3 > R_1$



(2017, 2019, 2023 Sample)

Q. A potential difference of 12 V is applied across a resistor. If the current through it is 3 A, calculate the resistance.

$$V = IR$$

$$R = \frac{V}{I} = \frac{12}{3} = 4 \Omega$$

FACTORS AFFECTING RESISTANCE

(i) Length of the conductor (l): $R \propto L$

Longer the wire $\rightarrow$ more resistance.

$$\downarrow R \propto \frac{1}{A} \uparrow$$

(ii) Inversely proportional to the area of cross-section: $R \propto 1/A$

Thicker wire (larger area) $\rightarrow$ less resistance.

(iii) Directly proportional to the temperature: $R \propto \text{Temperature}$

Metal $T \uparrow R \uparrow$
Alloy $T \uparrow R \uparrow$

- Metals like copper and aluminium have low resistance $\rightarrow$ good conductors.
- Materials like rubber and plastic have very high resistance $\rightarrow$ insulators.
- Some materials like semiconductors (silicon, germanium) have resistance values between conductors and insulators.

$$\rho \rightarrow R_{ho} \quad \uparrow T \uparrow R \downarrow$$

(iv) Depends on nature of material: (ρ)

- Different materials have different resistivity (ρ).
- Example: Copper has very low resistivity, nichrome has higher resistivity.

RESISTIVITY

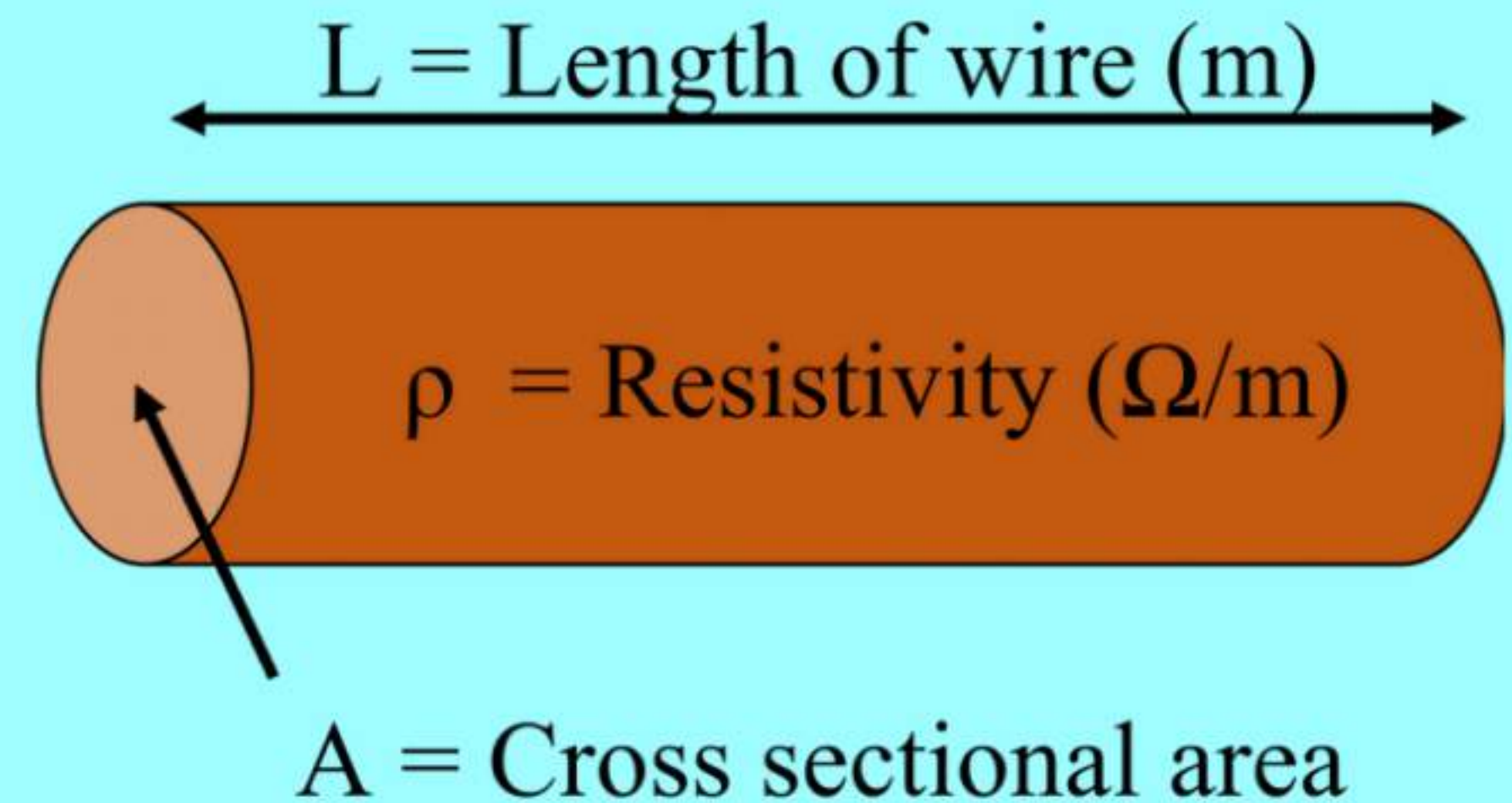
$$R = \rho L/A$$

$$\rho = RA/L$$

$$\rho = \Omega\text{m}$$

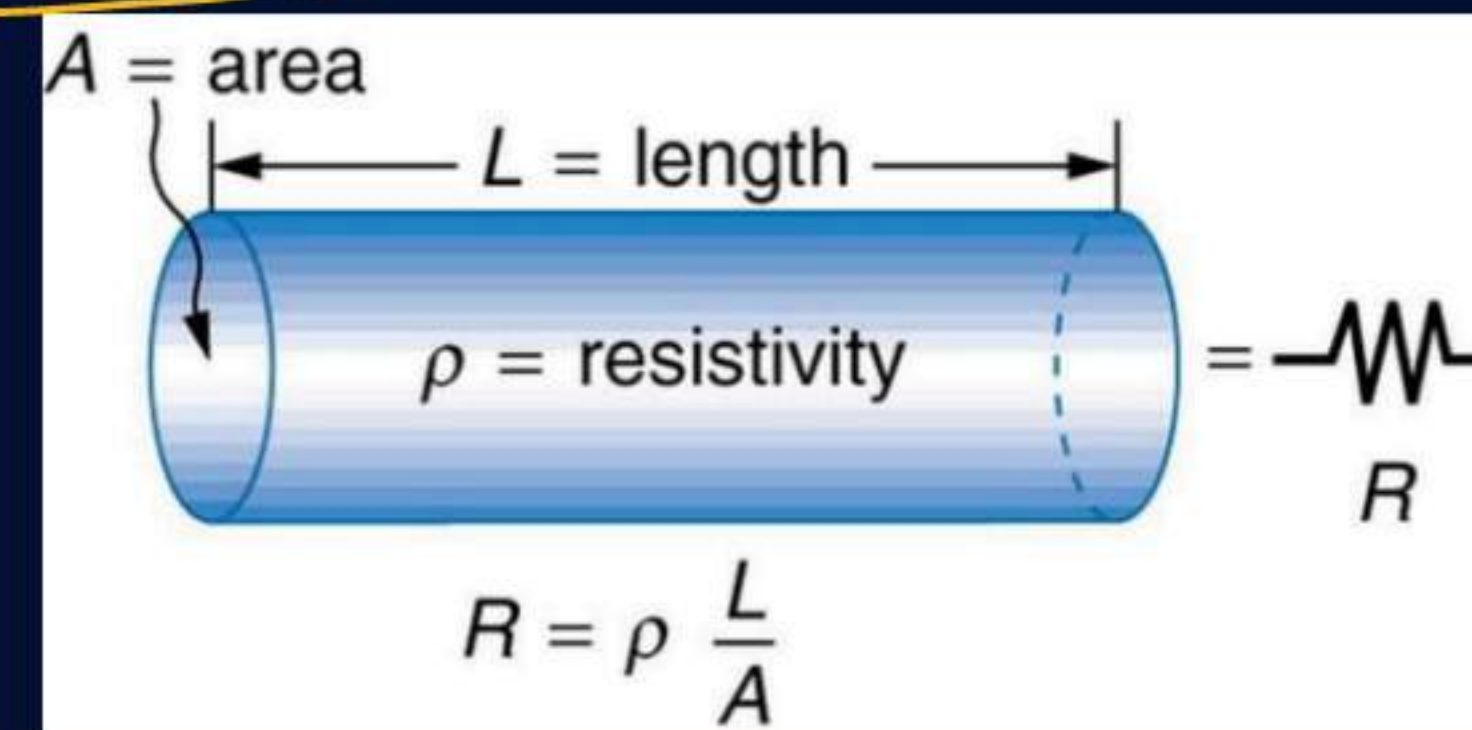
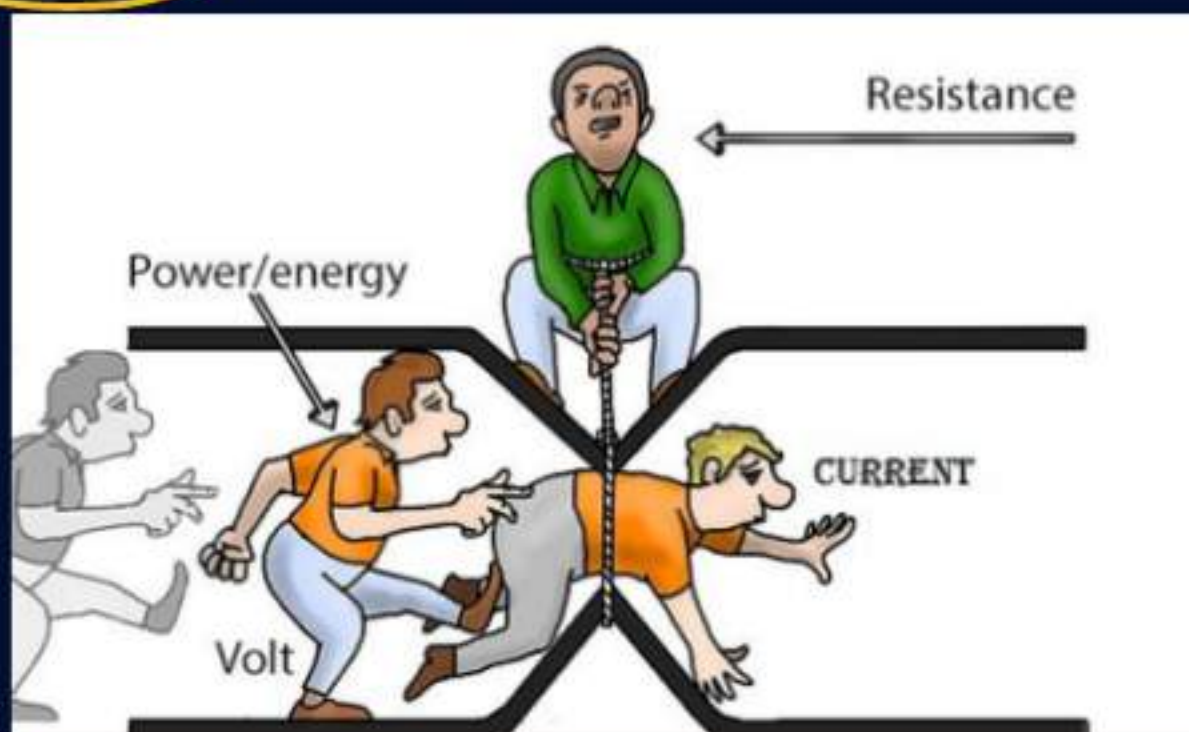
SI unit: Ohm metre ($\Omega\cdot\text{m}$)

- Resistance of a conductor is directly proportional to its length, inversely proportional to its area of cross-section, and depends on the material of the conductor. **It is denoted by ρ .**
- ***Resistivity does not change with change in length or area of cross-section but it changes with change in temperature.***
- Tungsten → filament of bulbs (high melting point).
- Copper, aluminium → transmission wires (low resistivity).



RESISTANCE V/S RESISTIVITY

Resistance (R)	Resistivity (ρ)
Opposition offered by a conductor to the flow of electric current.	The resistance of a conductor of unit length and unit cross-section.
Depends on length and area of cross-section of the conductor.	Independent of length and area of cross-section. Depends only on the nature of the material.
Formula: $R = \rho \times (L / A)$	Formula: $\rho = R \times (A / L)$
SI unit: ohm (Ω)	SI unit: ohm-metre ($\Omega \cdot m$)



$$R \propto L$$

$$R \propto \frac{1}{A}$$

~~Q~~

$$R = \frac{PL}{A}$$

$$\Omega = \frac{P \eta}{m k}$$

~~Q~~

$$P = \Omega m$$

2012, 2015, 2017, 2020 (Sample Paper), 2023 (Sample Paper)

Q. At 20°C , the resistivity of three materials A, B and C are as follows:

- $A = 1.6 \times 10^{-8} \Omega\text{m}$ ✓ Metal
- $B = 6.84 \times 10^{-8} \Omega\text{m}$ ✓ Alloy
- $C = 10^{12} \Omega\text{m}$ ✓

Identify them as metal, alloy and insulator. Which one is used to make heating elements?

(2011, 2017, 2020, 2025)

Q. The resistivity of a wire made of an alloy is generally:

- (a) Lower than that of its constituent metals.**
- (b) Higher than that of its constituent metals.**
- (c) Decreases with an increase in its area of cross-section.**
- (d) Increases with an increase in its length.**

(2016, 2019, 2022, 2025)

Q. A wire of length 'l' is gradually stretched so that its length increases to 3l. If its original resistance is R, then its new resistance will be:

(a) 3R

(b) 6R

~~(c) 9R~~

(d) 27R



Diagram of a wire of length l and resistance R . The formula $R = \frac{\rho L}{A}$ is written inside a circle.



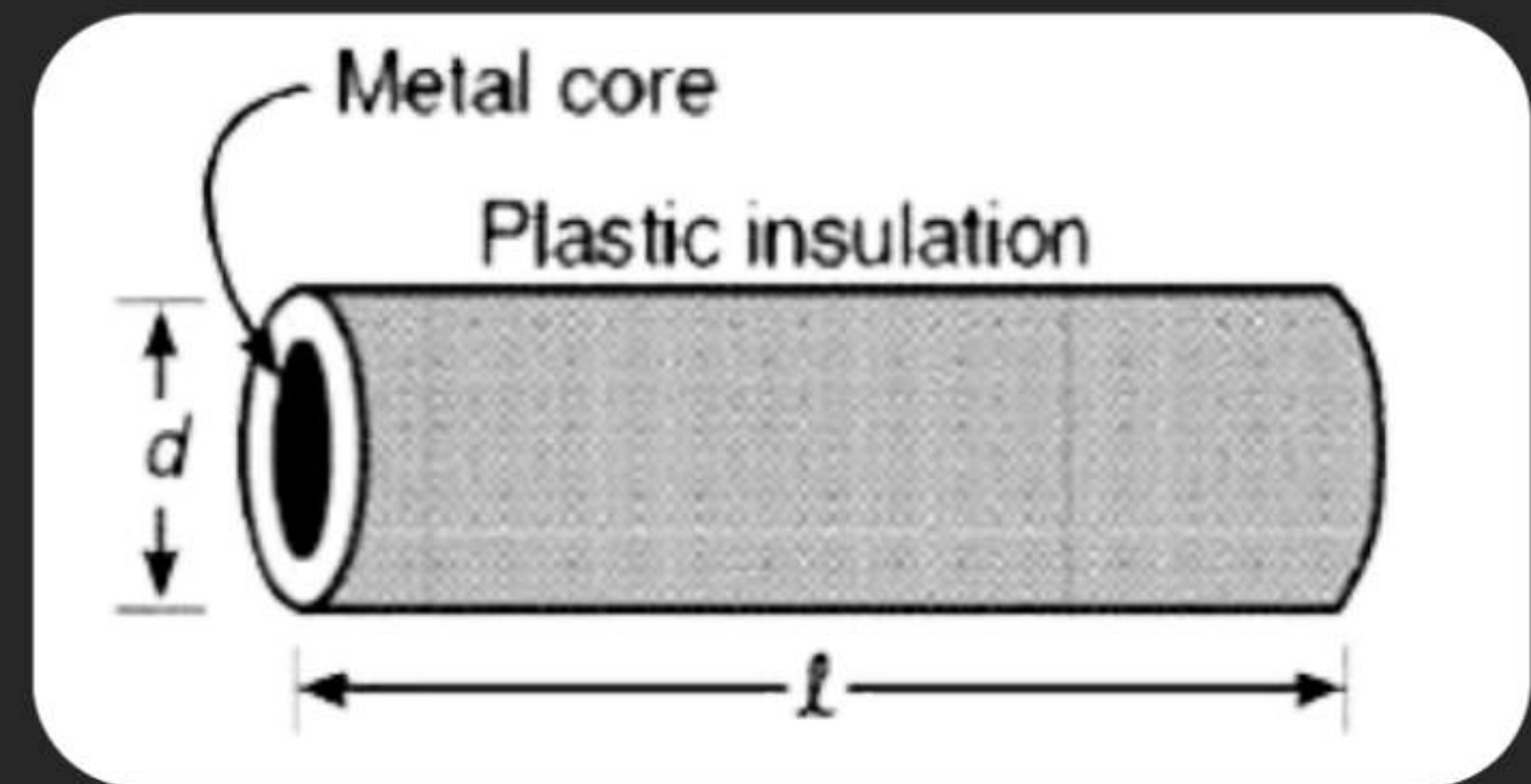
Diagram of a stretched wire of length $3l$ and cross-sectional area $\frac{A}{3}$. The formula $V = L \times A$ is written in a box.

$$R' = \frac{\rho 3L}{\frac{A}{3}} \times \frac{3}{A} = 9 \left(\frac{\rho L}{A} \right) = 9R$$
$$V = 3L \times \frac{A}{3}$$

2011, 2013, 2016, 2019

Q. Plastic insulation surrounds a wire having diameter d and length l as shown below. A decrease in the resistance of the wire would be produced by an increase in the:

- (a) length of the wire
- ☒ (b) thickness (diameter) of the wire
- (c) resistivity of the material
- (d) temperature of the wire



$$R \downarrow = \frac{\rho L}{A} \uparrow$$

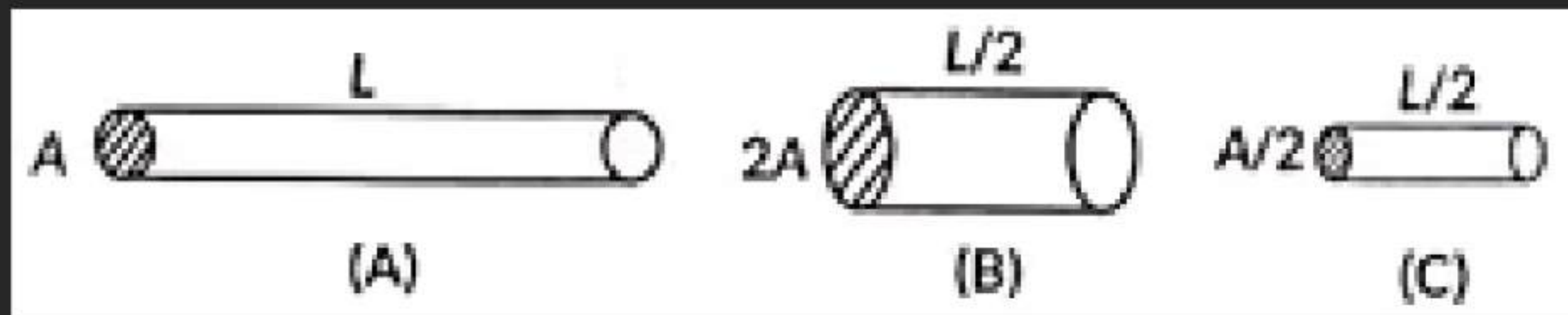
$$\frac{\pi r^2}{\text{}} = \pi \left(\frac{d}{2} \right)^2$$

$$d = 2r$$
$$r = d/2$$

(2022)

Q. In the following figure, three cylindrical conductors A, B and C are shown along with their lengths and areas of cross-section.

If these three conductors are made of the same material and R_A, R_B , and R_C be their respective resistances, then find (I) R_A/R_B , and (II) R_A/R_C .



$$R_A = \frac{\rho L}{A}$$

$$R_B = \frac{\rho \frac{L}{2}}{2A} = \frac{\rho L}{4A}$$

$$\frac{R_A}{R_B} = \frac{\frac{\rho L}{A}}{\frac{\rho L}{4A}} = \frac{4}{1} = 4:1$$

$$R_C = \frac{\rho \frac{L}{2}}{\frac{A}{2}} = \frac{\rho L}{A}$$

$$\frac{R_A}{R_C} = 1$$

(Sample Paper 2019)

Q. The resistance of a wire of length L is R . If the wire is compressed to half of its original length, what will be its new resistance?



The diagram shows two horizontal rectangles representing wires. The first rectangle on the left is labeled with length L above it and area A to its right. The second rectangle on the right is labeled with length $L/2$ above it and area $2A$ to its right. A curved arrow points from the first rectangle to the second, indicating a transformation.

$$R = \frac{\rho L}{A}$$
$$R' = \frac{\rho \frac{L}{2}}{2A} = \frac{1}{4} \left(\frac{\rho L}{A} \right)$$

CBSE 2024

Q. A copper wire has a diameter of 0.2 mm and resistivity of $1.6 \times 10^{-8} \Omega \text{ m}$. What will be the length of this wire to make its resistance 14Ω ? How much does the resistance change, if the diameter of the wire is doubled?

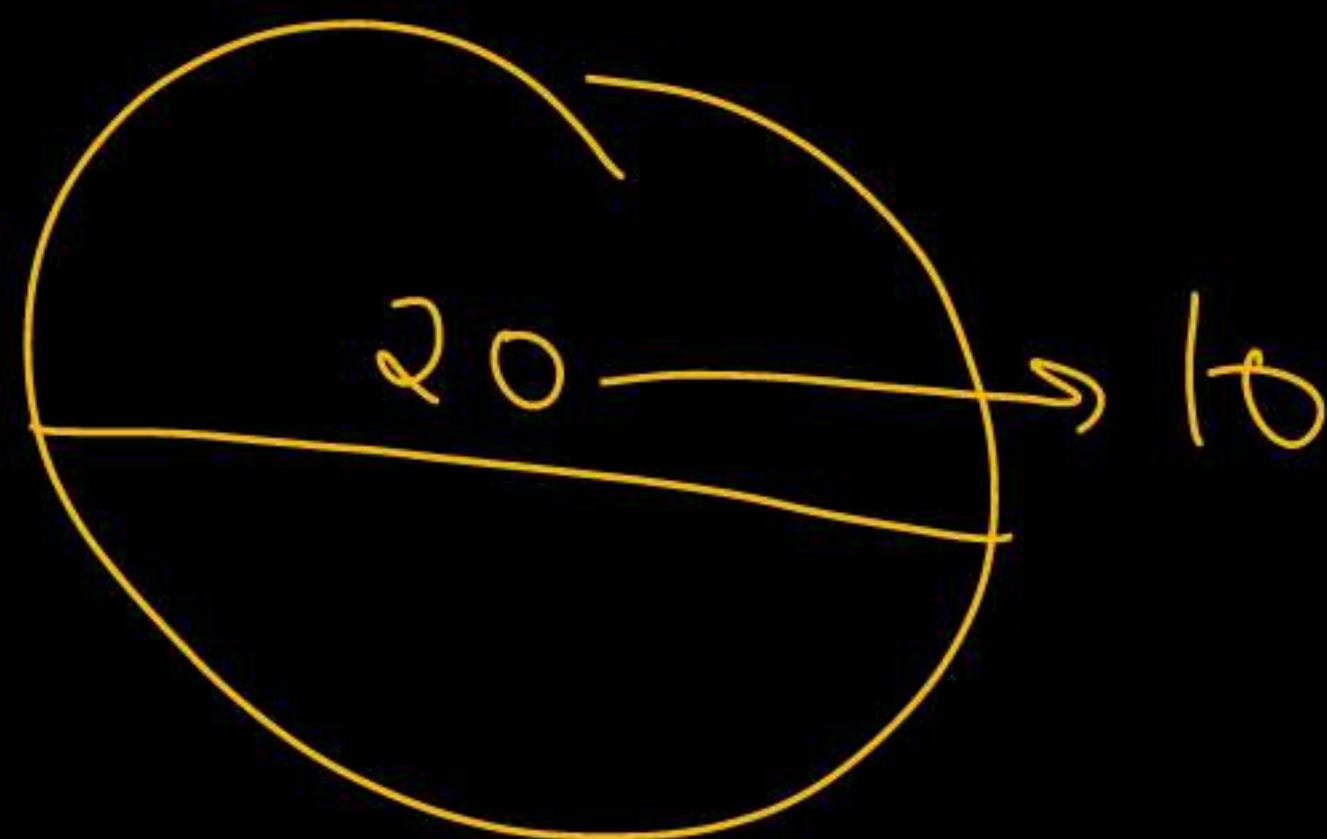
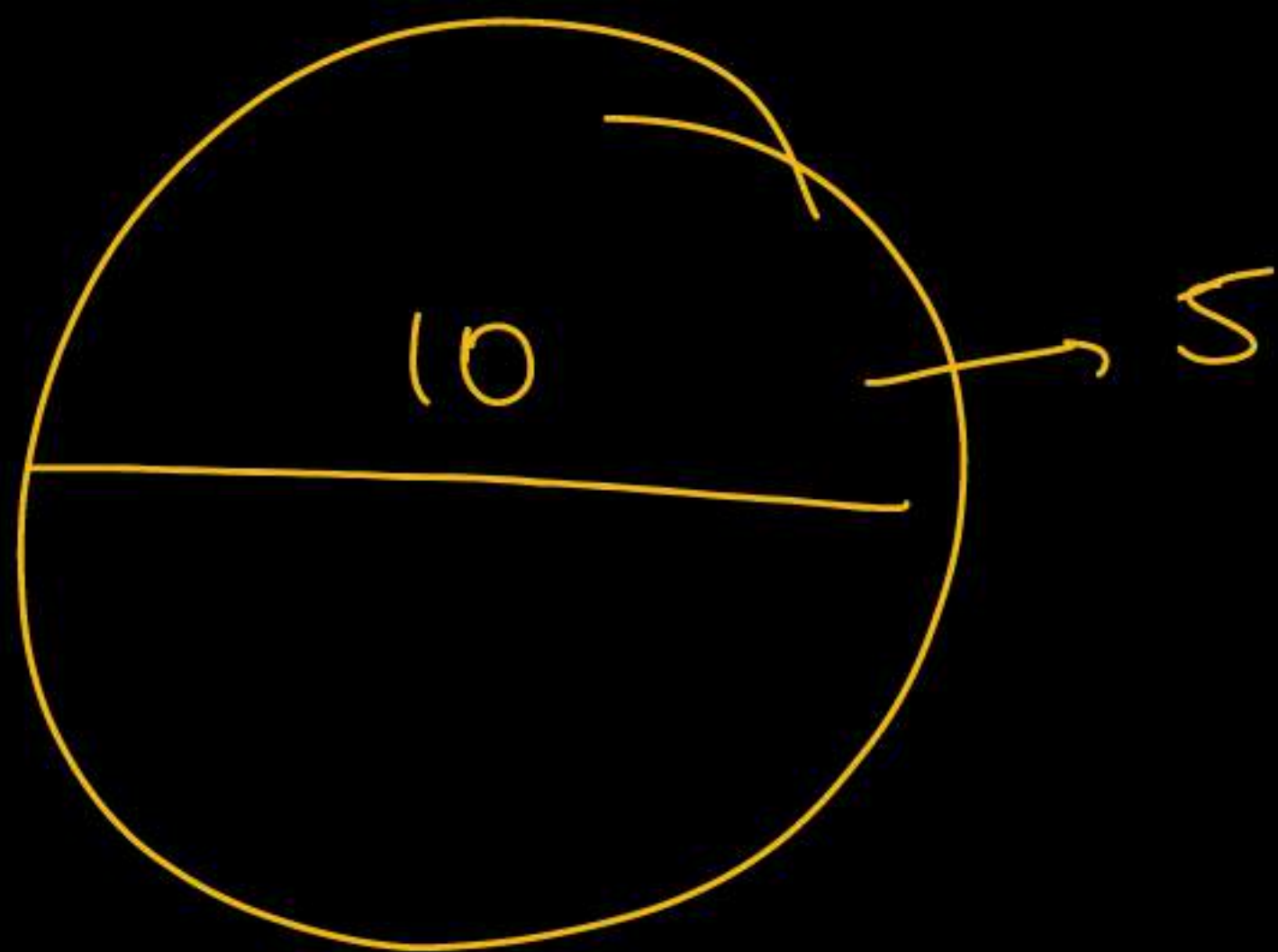
$$\rho = 1.6 \times 10^{-8}$$

$$R = 14 \checkmark$$

$$L = ?$$

$$A = \pi r^2$$

$$R = \frac{\rho L}{A}$$



$$R = \frac{PL}{\textcircled{A}} = \frac{PL}{\pi r^2}$$

↓

$$= \frac{PL}{\pi (2r)^2} = \frac{PL}{4\pi r^2}$$

SERIES COMBINATION:

When two or more resistors are connected end to end, the arrangement is called series combination.

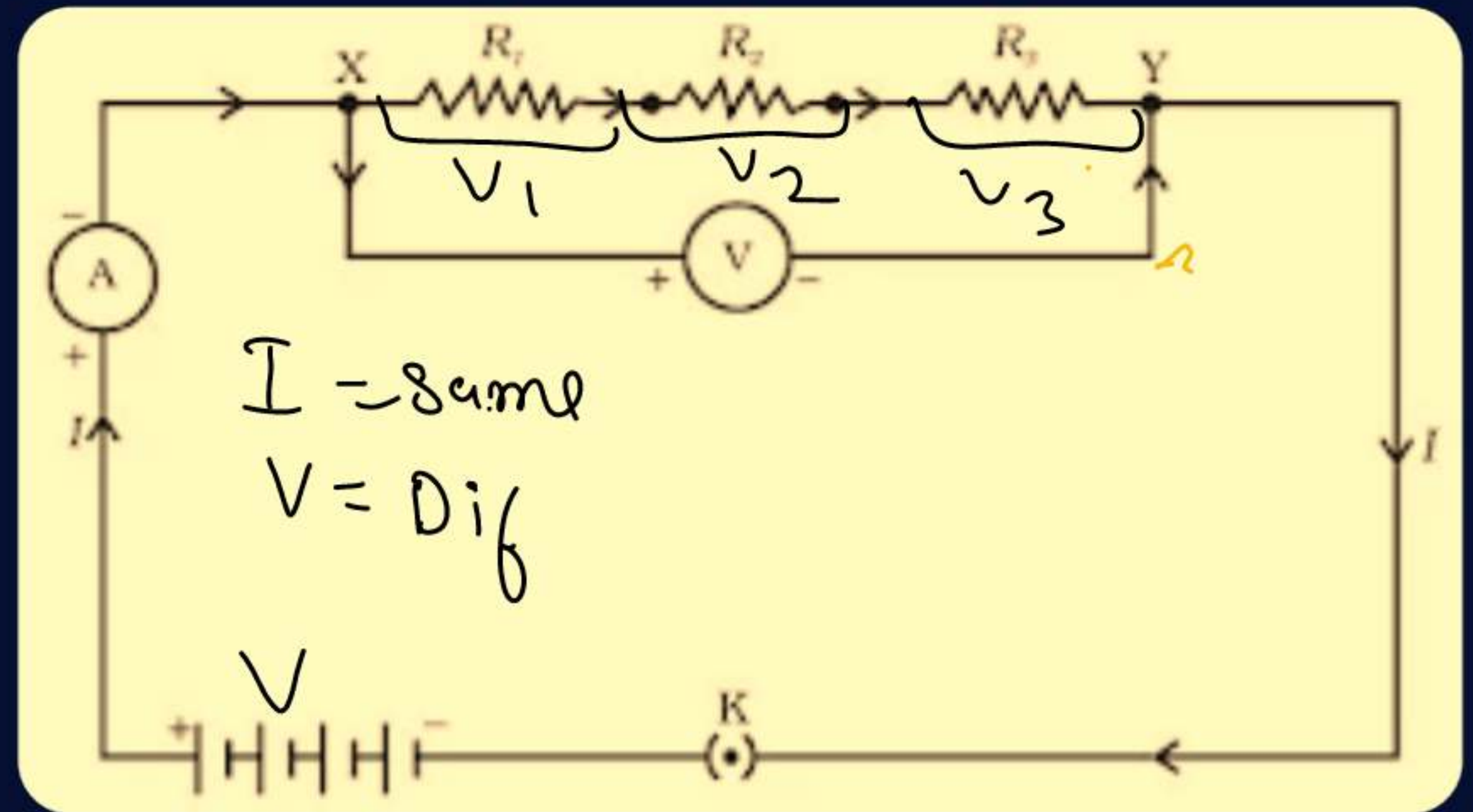
$$\underline{R_{net} = R_{eq}}$$

$$\textcircled{V} = V_1 + V_2 + V_3$$

$(V = IR)$

$$I R_{net} = I R_1 + I R_2 + I R_3$$

$$R_{net} = R_1 + R_2 + R_3$$



[2016, 2019, Sample Paper 2021, 2023 Board Exams]

Q. Calculate the equivalent resistance when resistors of $2\ \Omega$, $3\ \Omega$, and $6\ \Omega$ are connected in series.



$11\ \Omega$

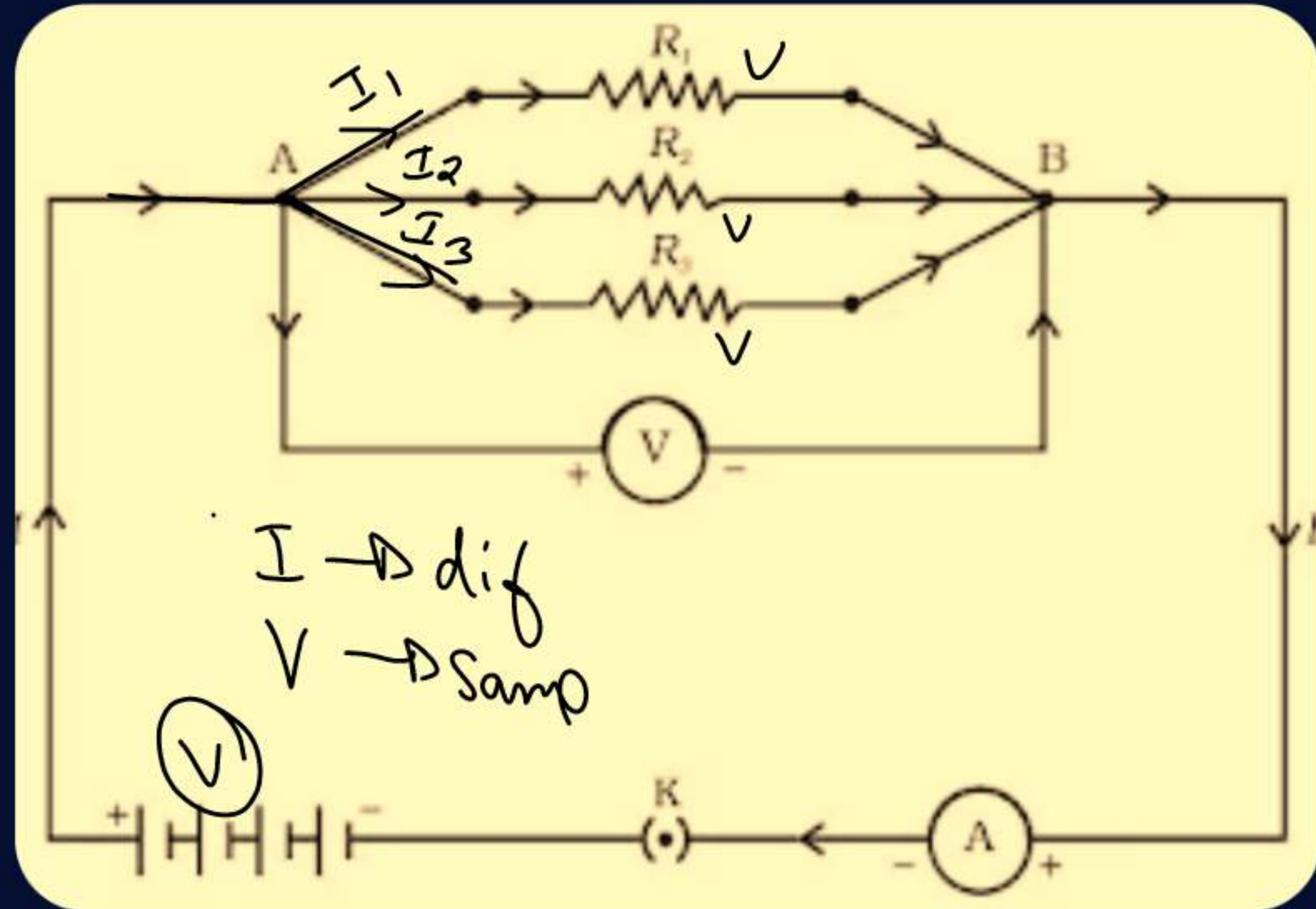
PARALLEL COMBINATION

When two or more resistors are connected across multiple branches.

$$\textcircled{I} = I_1 + I_2 + I_3$$
$$\left[I = \frac{V}{R} \right]$$

$$\frac{V}{R_{eq}} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$\boxed{\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$



[2018]

Q. Three resistors are shown in the figure. The equivalent resistance of the combination is-

a) 3Ω

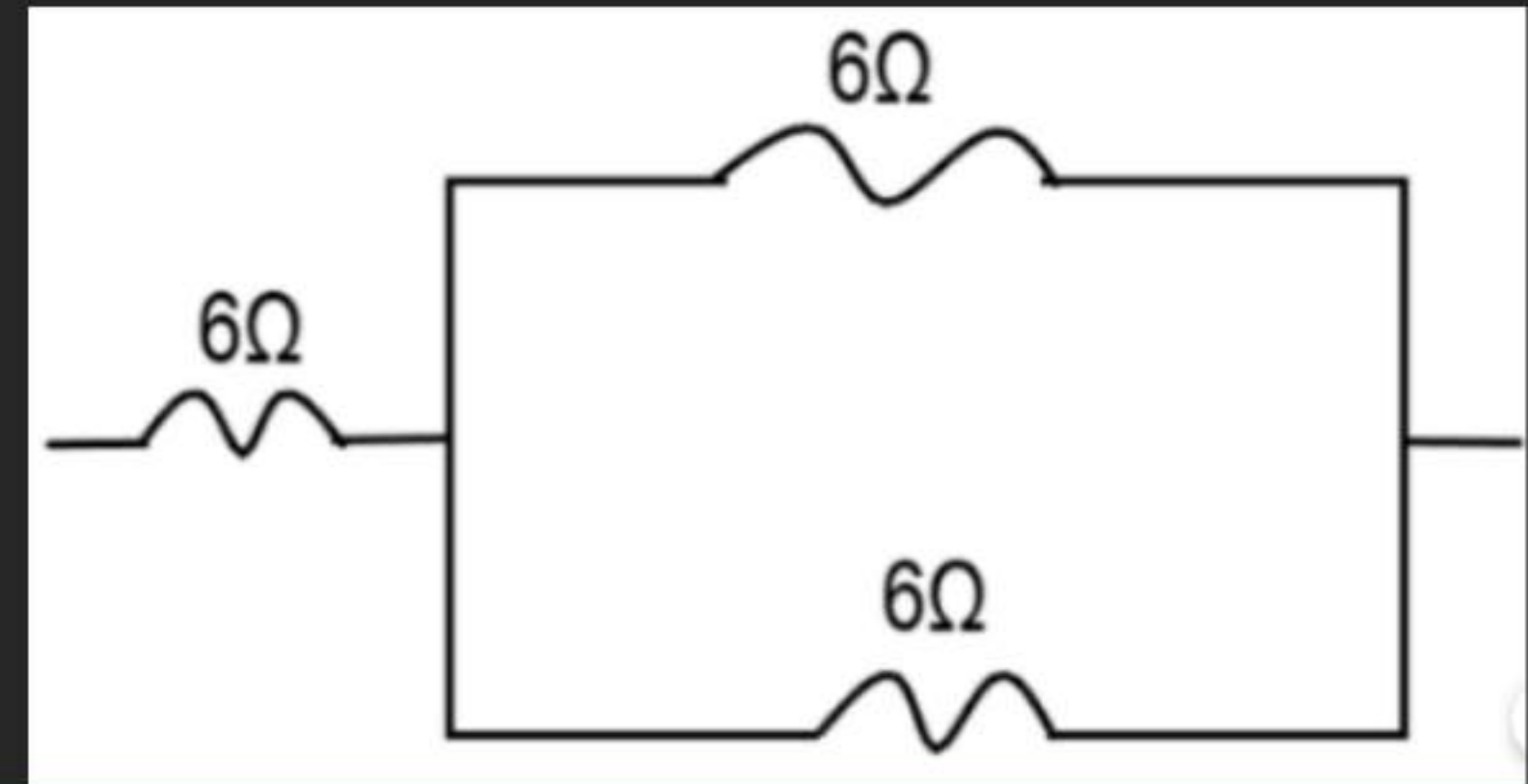
b) 6Ω

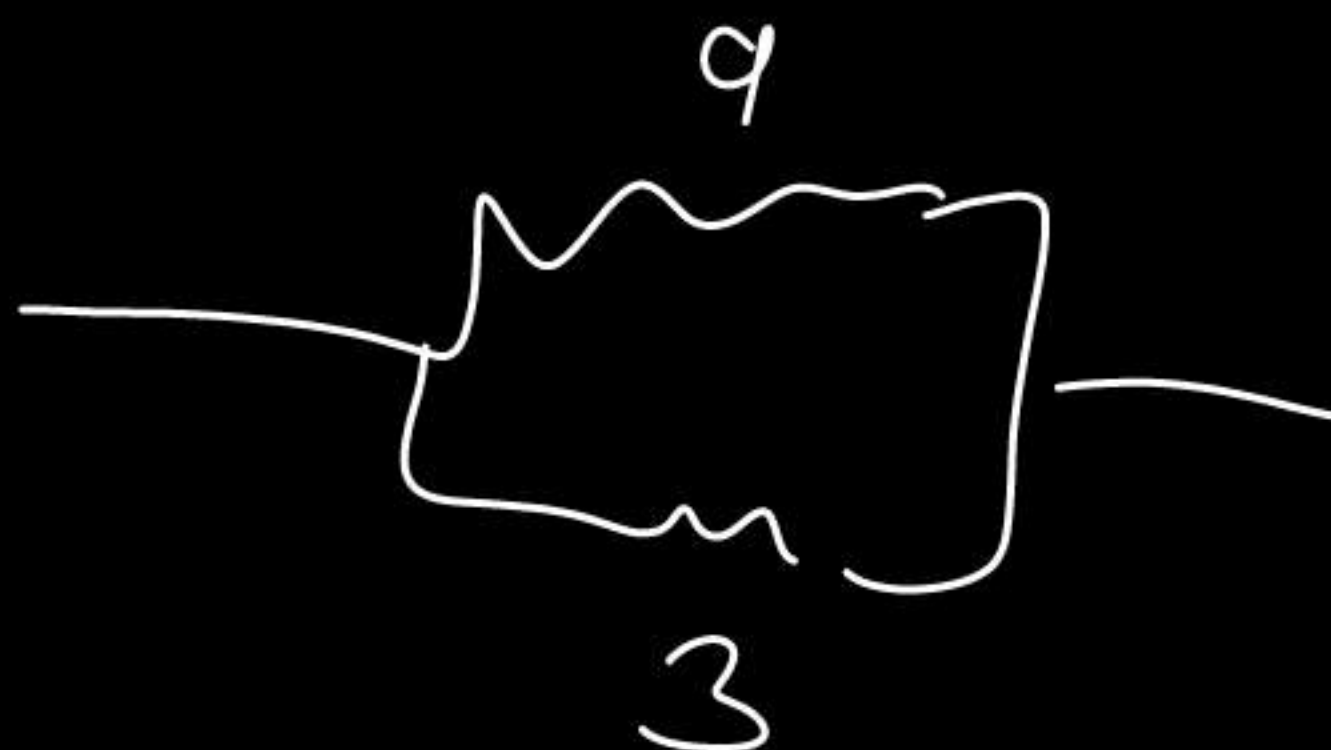
c) 9Ω

d) 7Ω

$$\frac{1}{R_{eq}} = \frac{1}{6} + \frac{1}{6}$$
$$\frac{1}{R_{eq}} = \frac{2}{6} = \frac{1}{3}$$

$$R_{eq} = 3\Omega$$

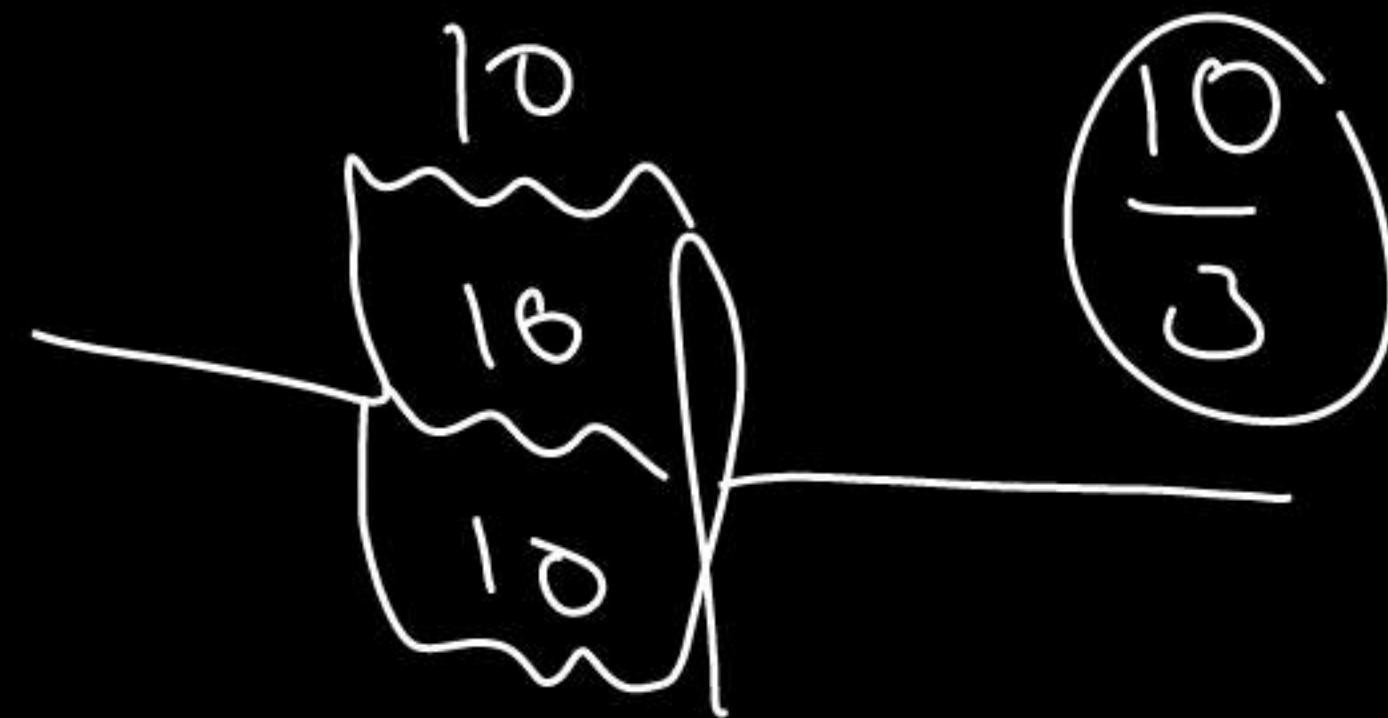
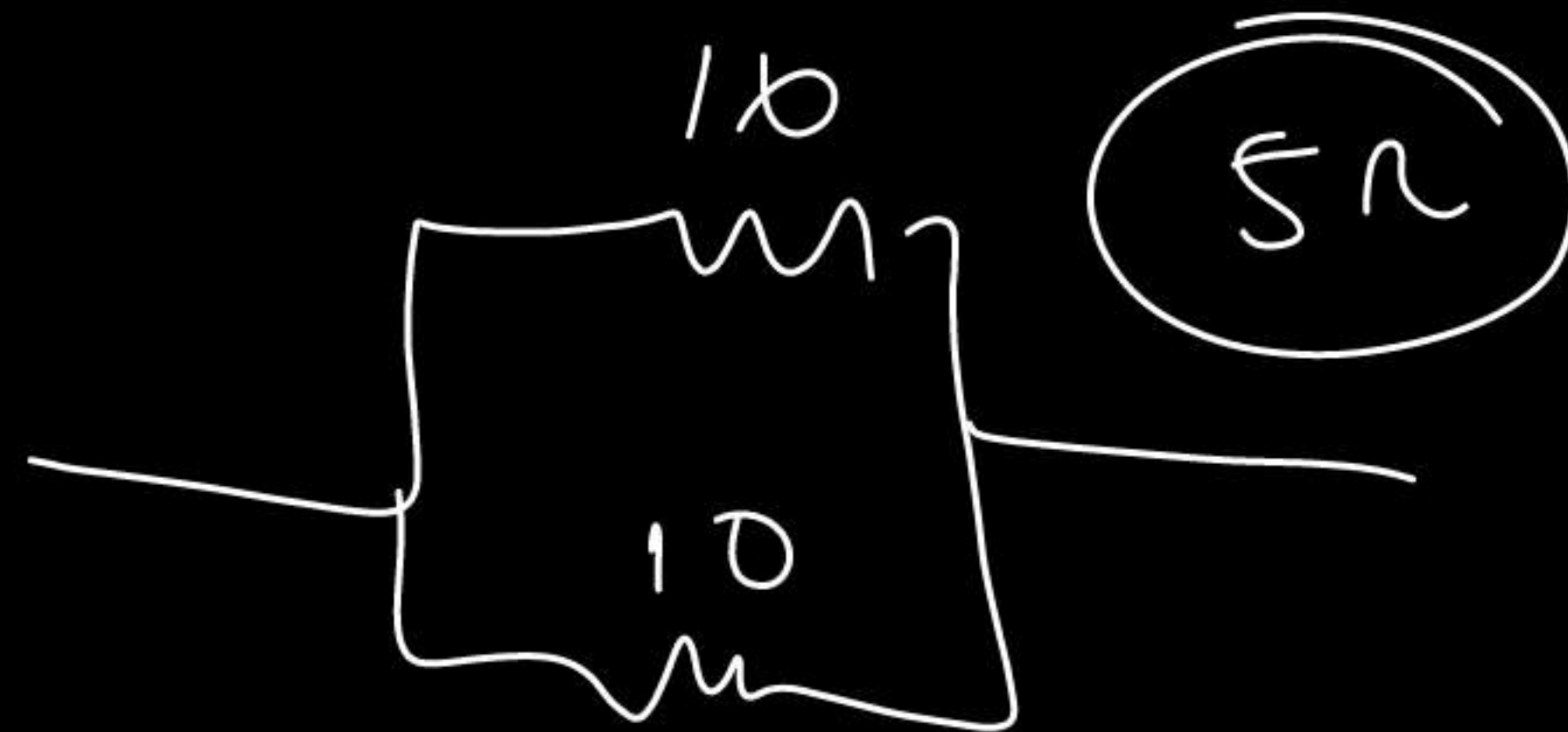




$$R_{eq} = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{9 \times 3}{9 + 3} = \frac{27}{12} = \frac{9}{4}$$

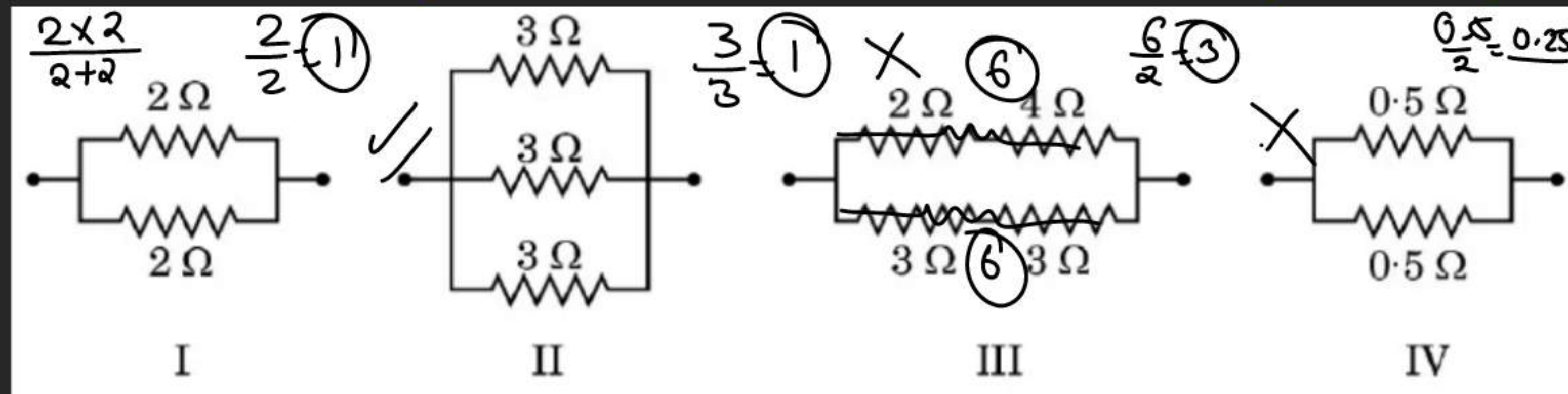
$$\frac{1}{9} + \frac{1}{3} = \frac{1+3}{9} = \frac{4}{9}$$

←



[2024]

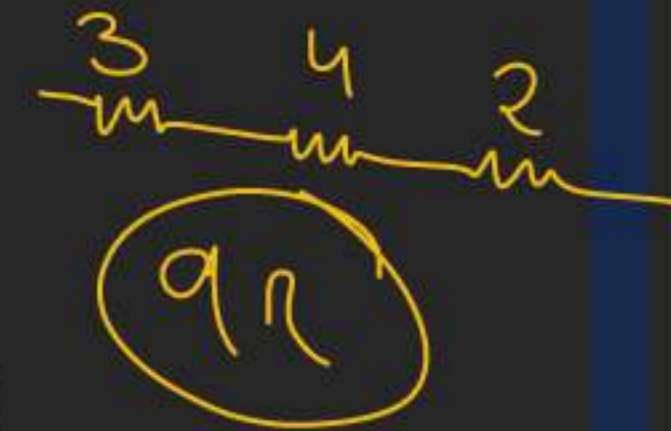
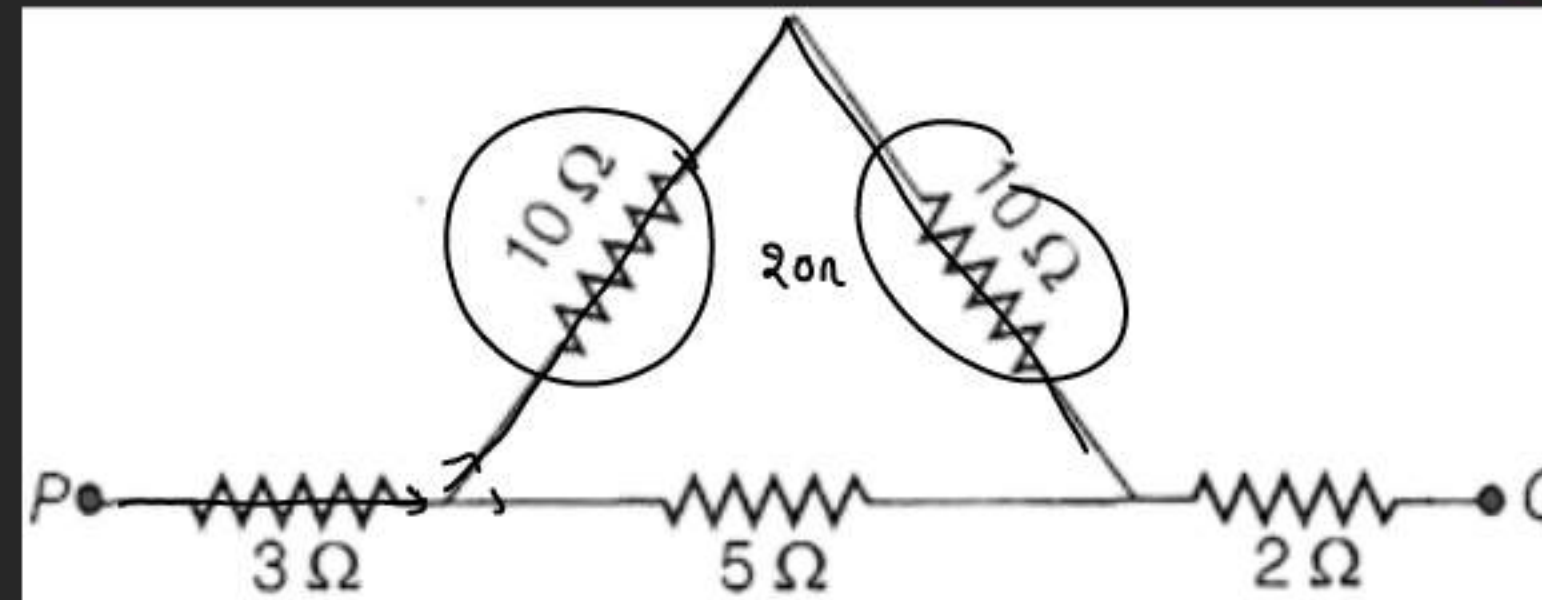
Q. Consider the following combinations of resistors: (1 Mark)



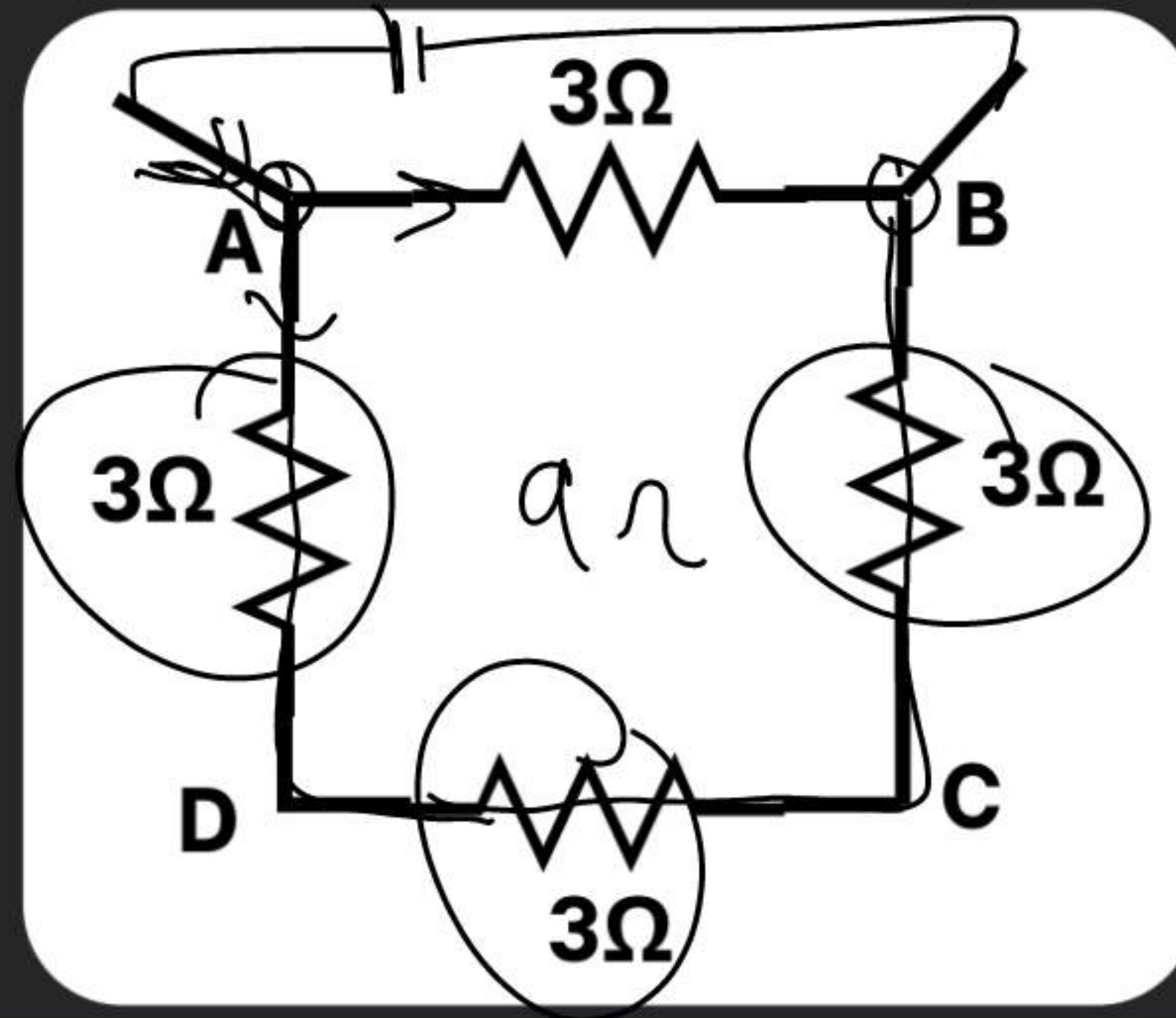
The combinations having equivalent resistance 1 is/are:

- (a) I and IV
- (b) Only IV
- (c) I and II
- (d) I, II and III

Q. Calculate the equivalent resistance



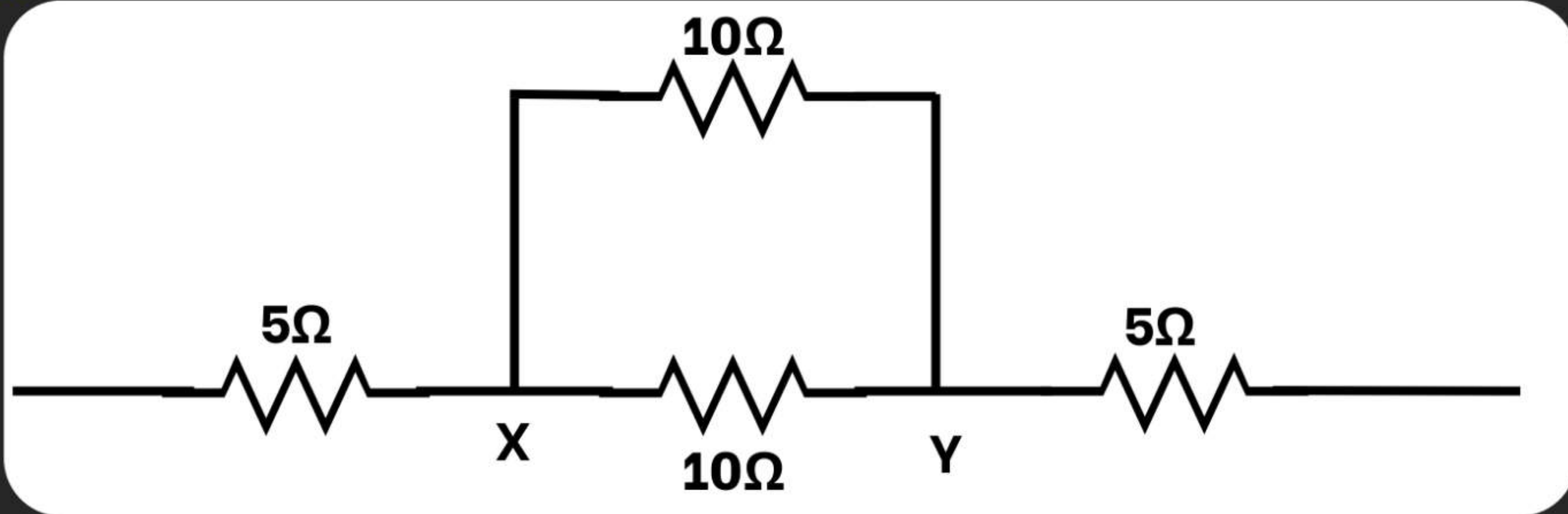
Q. In the given circuit, calculate the equivalent resistance between points A and B:



Handwritten calculation for equivalent resistance:

$$R_{\text{net}} = \frac{27}{\frac{1}{2}} = \left(\frac{9}{4}\right)\Omega$$

Q. Find the equivalent resistance between points X and Y in the following circuit:

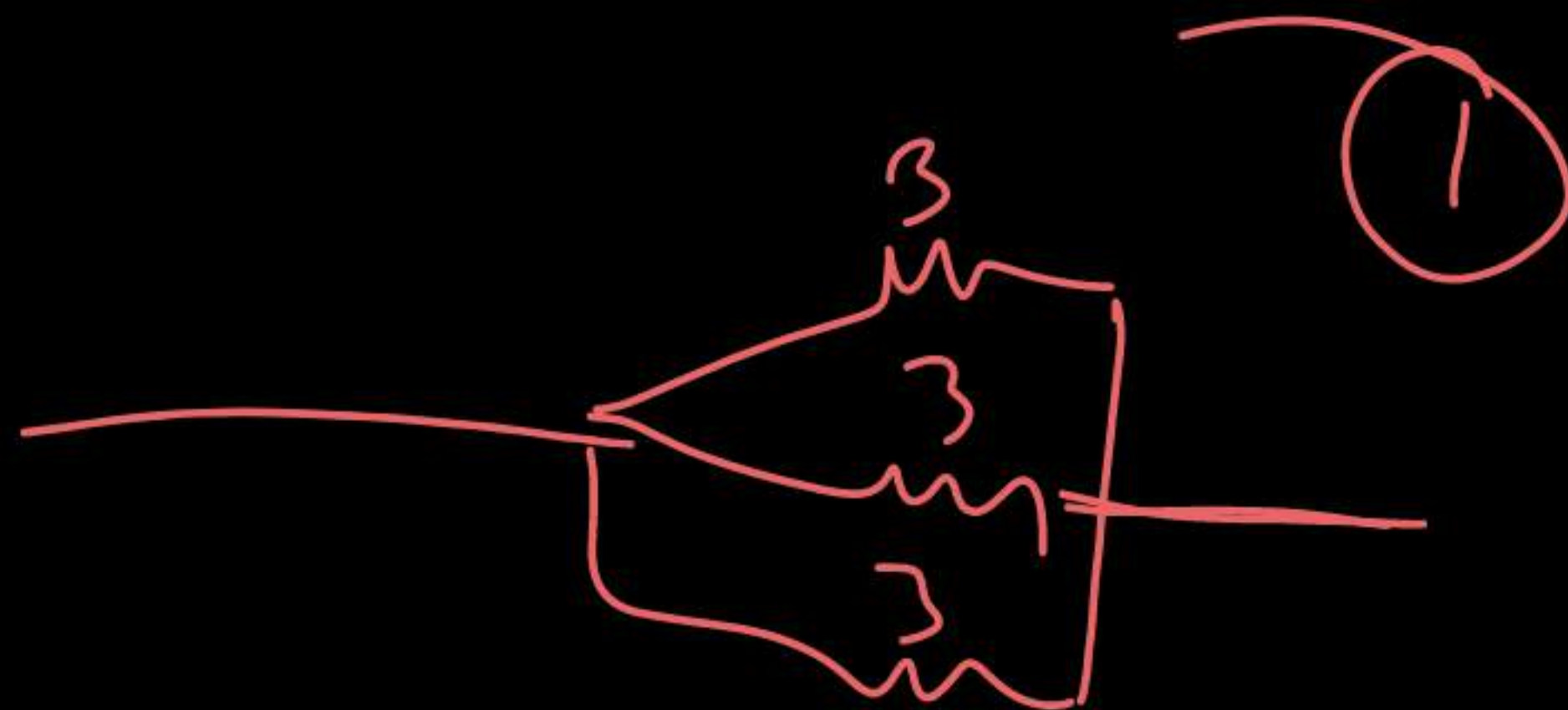


[2016, 2019, 2020 Sample Paper]

Q. What is (a) the highest total resistance that can be secured by a combination of four resistance of $4\Omega, 8\Omega, 12\Omega$ and 24Ω ? $= 48\Omega$

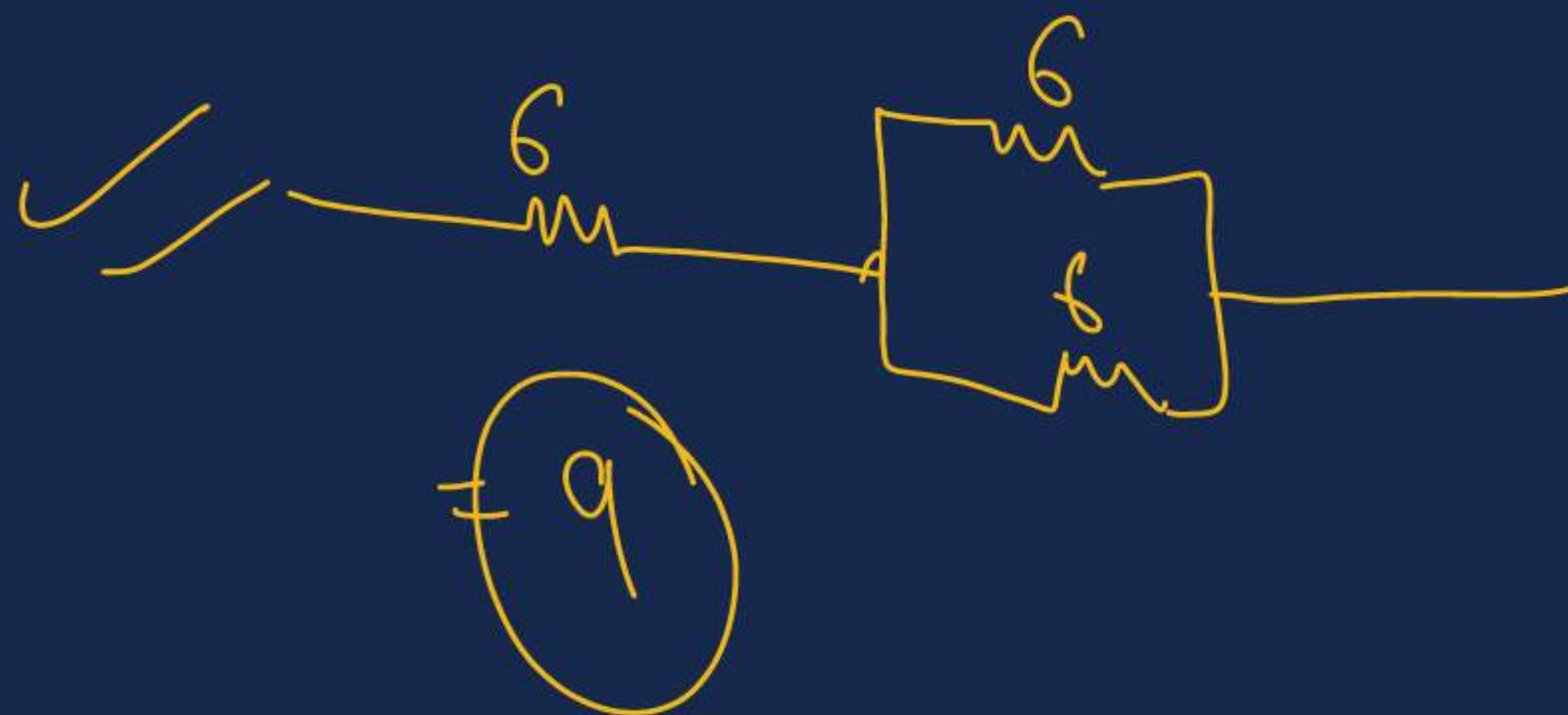
b) The lowest total resistance that can be secured by a combination of four resistance of $4\Omega, 8\Omega, 12\Omega$ and 24Ω ?

$$\begin{aligned}\frac{1}{R} &= \frac{1}{4} + \frac{1}{8} + \frac{1}{12} + \frac{1}{24} \\ &= \frac{6+3+2+1}{24} = \frac{12}{24} = \frac{1}{2} \Rightarrow 2\Omega\end{aligned}$$



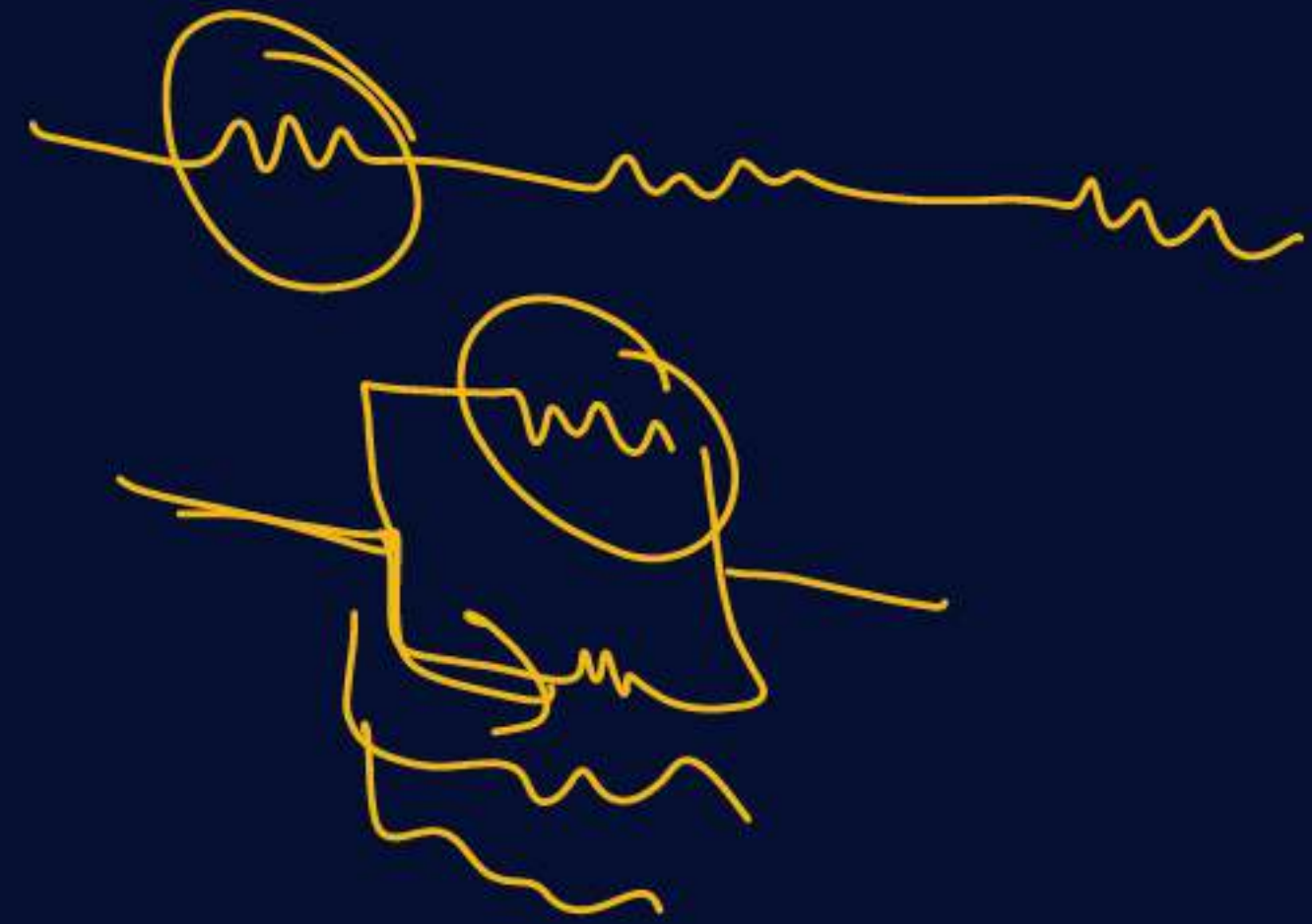
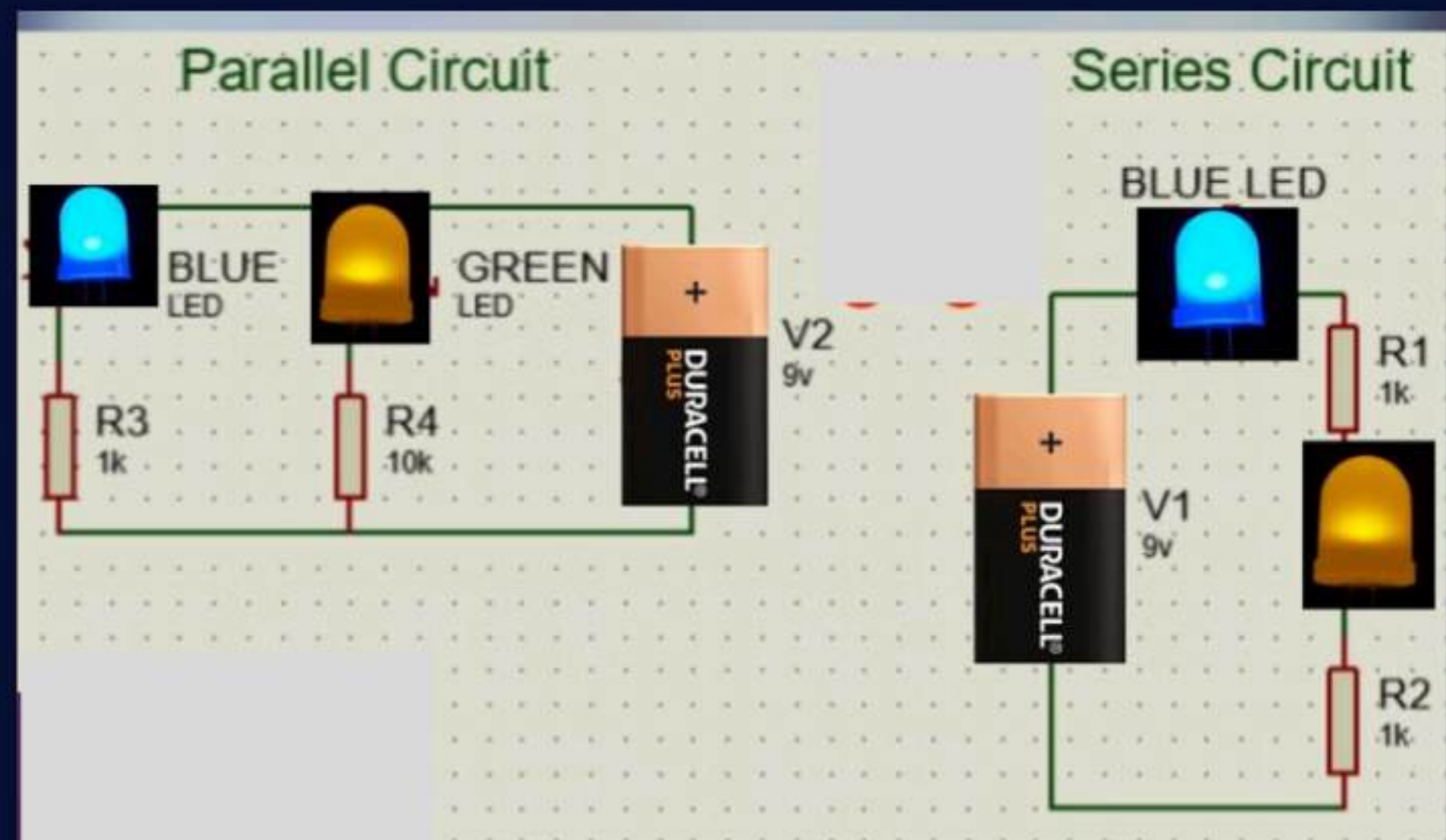
①
2015, 2018

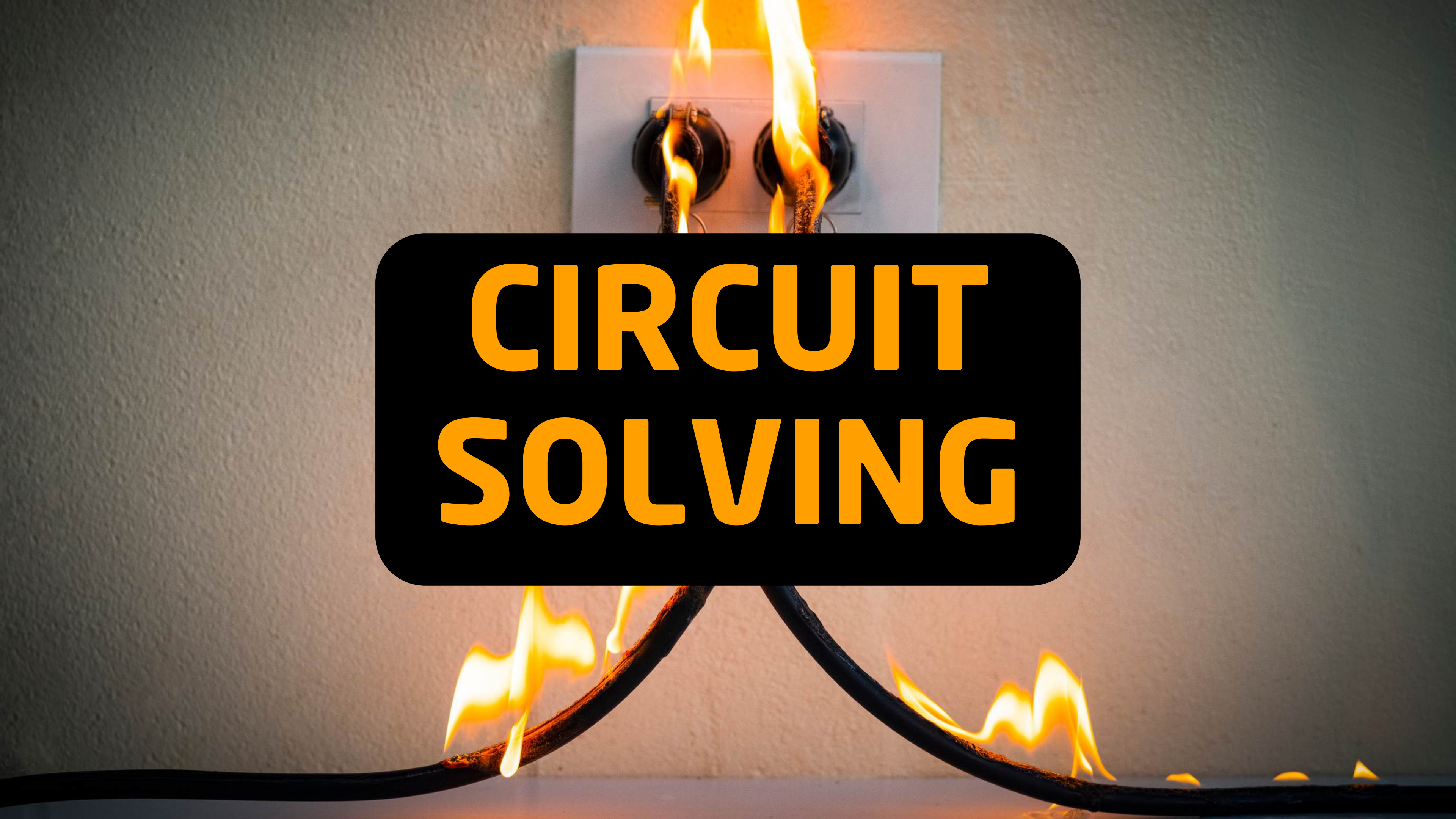
Q. How would you connect three resistors each of resistance $6\ \Omega$ so that the effective resistance of the combination is (a) $9\ \Omega$, (b) $2\ \Omega$



ADVANTAGE OF PARALLEL OVER SERIES COMBINATION

- **Independent Functioning:** Each device works independently. If one fails, others continue to work.
- **Same Voltage:** All components receive the same voltage as the source.
- **Reduced Resistance:** Adding more devices decreases the total resistance, allowing efficient current flow.
- **Easy to Add Devices:** New devices can be added without affecting others.

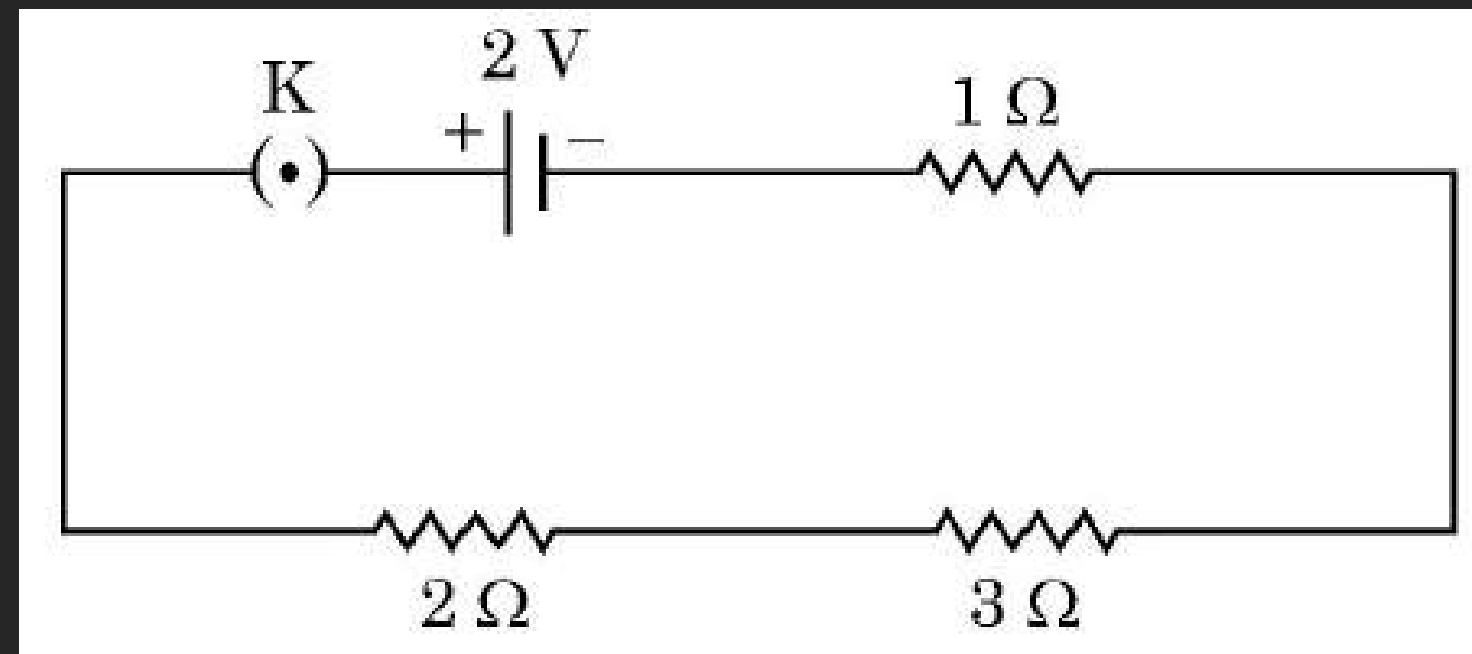


A photograph of a wall outlet with two lit candles in the sockets and two lit candles in holders below, with the text "CIRCUIT SOLVING" overlaid. The background is a textured, light-colored wall. The candles are lit, and the flames are visible. The text is in a bold, orange, sans-serif font, centered on a black rectangular background with rounded corners.

CIRCUIT SOLVING

[2024]

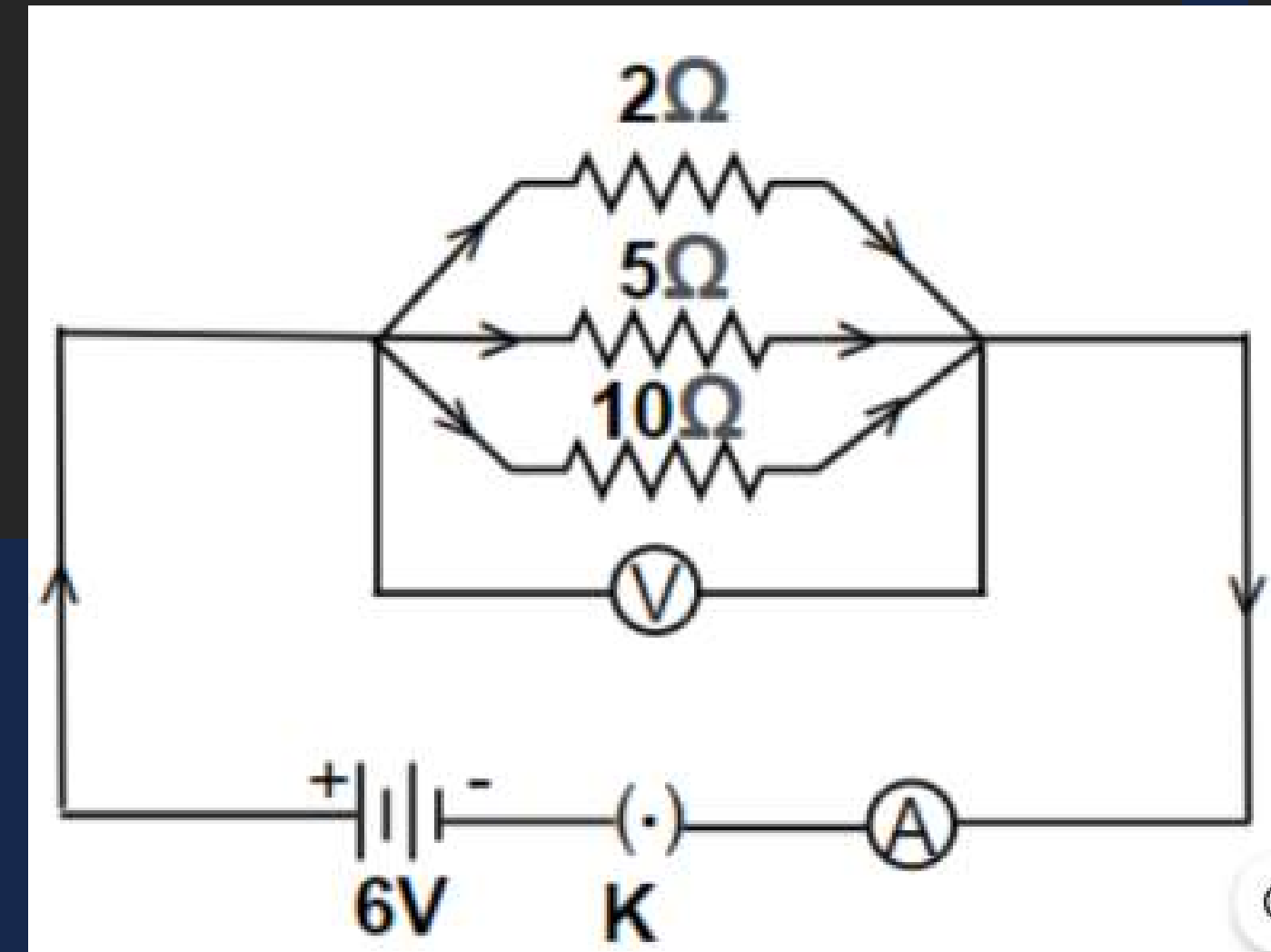
Q. Use Ohm's law to determine the potential difference across the $3\ \Omega$ resistor in the circuit shown in the following diagram when the key is closed.



2015, 2016, 2019, 2023, Sample Papers 2020, 2022.

Q. In the circuit diagram given here Calculate-

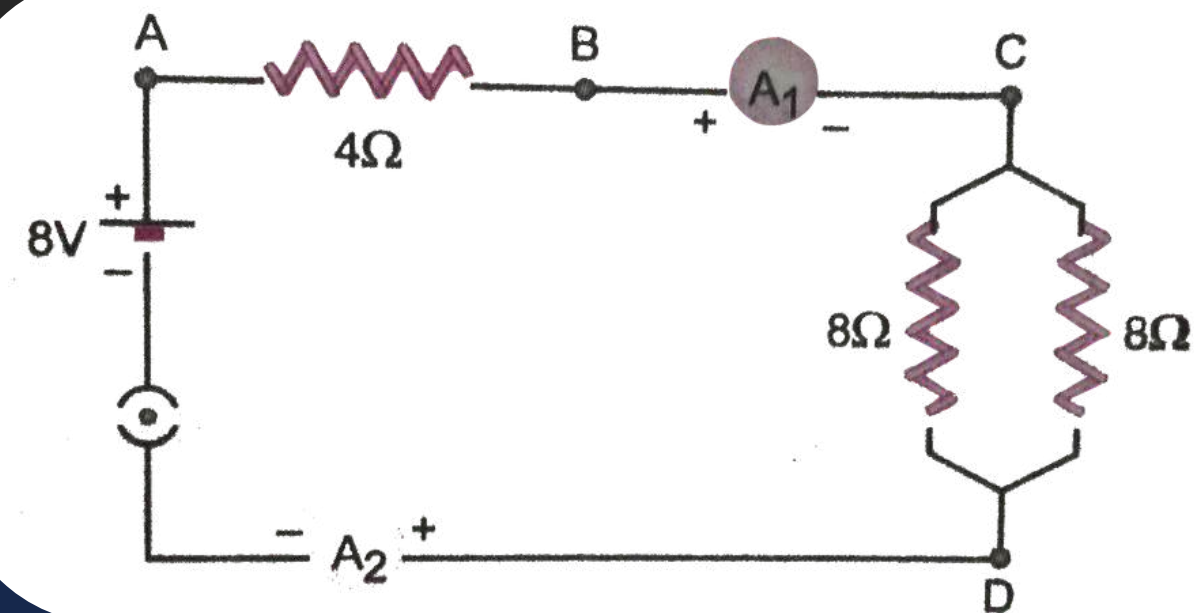
- a) The total effective resistance**
- b) The total current**
- c) The current through each resistor.**



[2018,2025 Sample Paper]

Q. Find out the following in the electric circuit given in the figure-

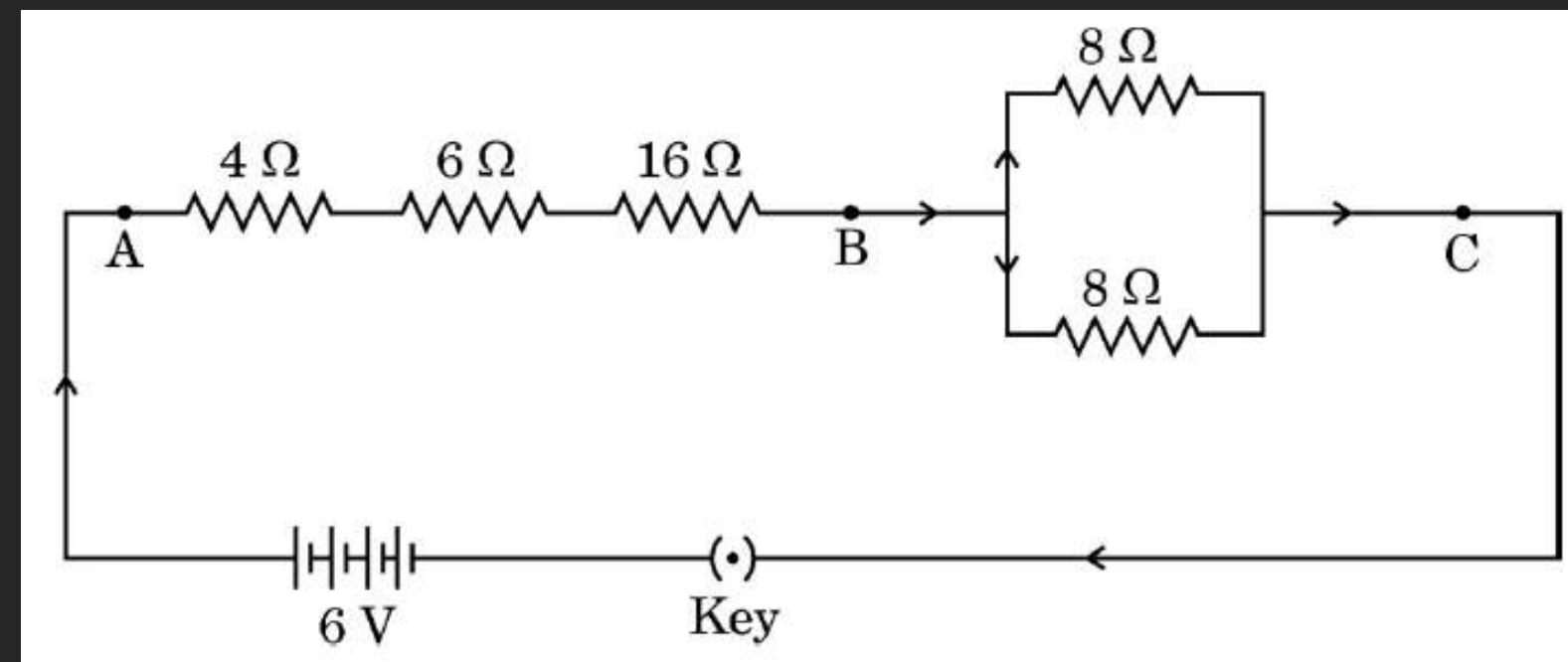
- (i) Effective resistance of two 8 ohm resistors in**
- (a) the equivalent resistance of the circuit,**
- (b) the current in the circuit and the readings of A_1 and A_2 ,**
- (c) the potential difference across the $4\ \Omega$ resistor and across each $8\ \Omega$ resistor.**



Q. The following questions are source-based/case-based questions. Read the case carefully and answer the questions that follow. Study the following circuit:

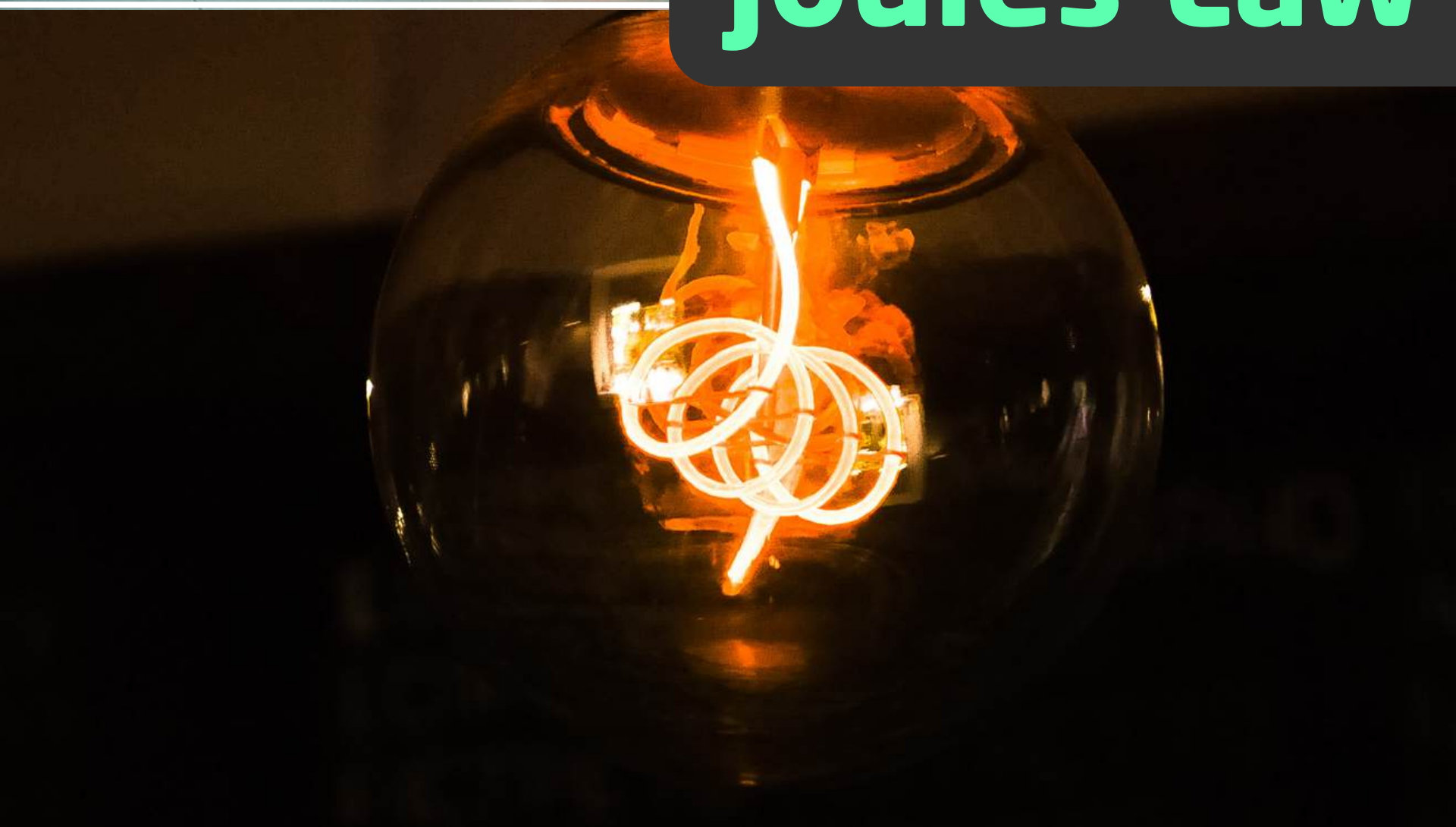
On the basis of this circuit, answer the following questions:

- (a) Find the value of total resistance between points A and B.**
- (b) Find the resistance between the points B and C.**
- (c) (i) Calculate the current drawn from the battery, when the key is closed.**





Joules Law of Heating



Joules Law of Heating

Joule's law of heating states that the heat produced in a conductor is directly proportional to:

- (i) the square of the current (I^2),*
- (ii) the resistance of the conductor (R), and*
- (iii) the time for which the current is passed (t).*

Work done in moving charge Q through potential difference V :

$$W = VQ$$

In time t , charge $Q=It$.

$$W = V \times It$$

This energy appears as heat (H) in the resistor:

$$H = VIt$$

Using Ohm's law ($V=IR$):

$$\mathbf{H = I^2 R t}$$

$$\mathbf{H = I^2 R t}$$


2012, 2015, 2017, 2019, 2020 (Sample), 2023 (Board)

Q. A resistor of resistance $10\ \Omega$ is connected across a battery of $12\ \text{V}$. Calculate the heat produced in the resistor in 5 minutes.

Electric Bulb

- **An electric bulb contains a thick tungsten filament having a high melting point of about 3380°C .**
- **The filament is enclosed in a glass bulb which is either evacuated (vacuum) or filled with an inert gas such as argon or nitrogen to prevent the filament from burning.**



APPLICATION OF HEATING EFFECT OF ELECTRIC CURRENT

Electric Fuse

- An electric fuse is a safety device used in electric circuits to protect appliances from damage caused by excess current.
- **Working:** It consists of a thin wire made of an alloy of lead (Pb) and tin (Sn), which has a low melting point.
- Connected in series with the circuit.
- **Encasing:** Fuse wire is enclosed in a porcelain cartridge with metal ends.
- **Rating:** 1 A, 2 A, 3 A, 5 A, 10 A etc. (domestic use).
- If current exceeds the safe limit → wire melts due to heating effect → circuit breaks → appliances are saved.



Electric Heater/ Heating element

- The heating element is made of nichrome wire (alloy of nickel and chromium).
- Nichrome is used because:
 - a. High resistivity
 - b. High melting point
 - c. Does not oxidise quickly at high temperature.



Elements used :

- Filament of the bulb → Tungsten (high melting point $\sim 3380^{\circ}\text{C}$)
- Connecting wires → Copper (good conductor of electricity)
- Heating elements → Nichrome (alloy of Nickel and Chromium, high resistivity, high melting point)
- • Fuse wire → Lead–Tin (Pb–Sn) alloy (low melting point, high resistivity)

Q. Why are alloys like nichrome used for making heating devices such as electric heater coils and toasters?

2015, 2018, 2019, 2022 Boards; 2024 Sample

Alloys such as nichrome have high resistivity and high melting point. They do not oxidise or burn easily at high temperature, so they can withstand heating for a long time without damage.

Q. Give reasons for the following:

(a) Why is an electric heater coil made of nichrome and not copper?

2015, 2018, 2019, 2022 (Board), 2024 (Sample)

(b) Why is tungsten used in electric bulb filament?

2016, 2017, 2020, 2021 (Sample), 2023 (Sample)

(c) Why is a fuse wire made of a material with low melting point

2015, 2018, 2020 (Board), 2022 (Sample)

(a) Nichrome has high resistivity and high melting point, so it produces more heat and does not oxidise easily.

(b) Tungsten has a very high melting point ($\sim 3380^{\circ}\text{C}$) and can glow white-hot without melting.

(c) Fuse wire melts quickly when current exceeds the safe limit, thus breaking the circuit and protecting appliances.

2015 (Sample), 2017, 2019, 2021, 2023, 2024 (Board)

Q. Two conducting wires of the same material and of equal lengths and equal diameters are first connected in series and then parallel in a circuit across the same potential difference. The ratio of heat produced in series and parallel combination would be:

- (a) 1:2**
- (b) 2:1**
- (c) 1:4**
- (d) 4:1**

H.W.

Q. Compute the heat generated while transferring 96000 coulomb of charge in two hours through a potential difference of 40 V.

Given: Charge (Q) = 96000 C, Potential difference (V) = 40 V, Time (t) = 2 hours = $2 \times 3600 = 7200$ s

Step 1: Current

$$I = Q / t$$

$$I = 96000 / 7200$$

$$I = 13.33 \text{ A}$$

Step 2: Heat generated (Joule's Law)

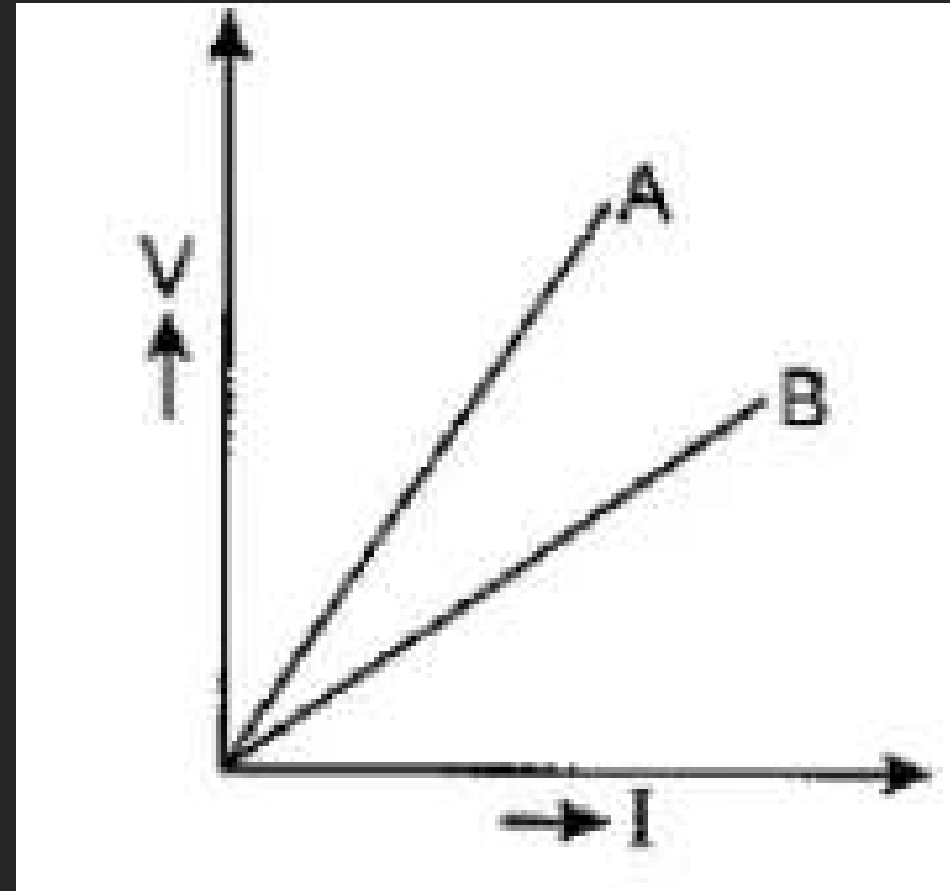
$$H = V \times I \times t$$

$$H = 40 \times 13.33 \times 7200$$

$$H = 3.84 \times 10^7 \text{ Js}$$

2016

Q. $V - I$ graphs for two wires A and B are shown in the Fig. 12.34. Both of them are connected in series to a battery. Which of the two will produce more heat per unit time? Give justification for your answer. (2016)



ELECTRICAL POWER

It is denoted by p .

Its S.I. unit is Watt (W) or J/s.

- The rate at which electric energy is consumed or dissipated in a circuit.
- It represents the rate of doing electrical work.

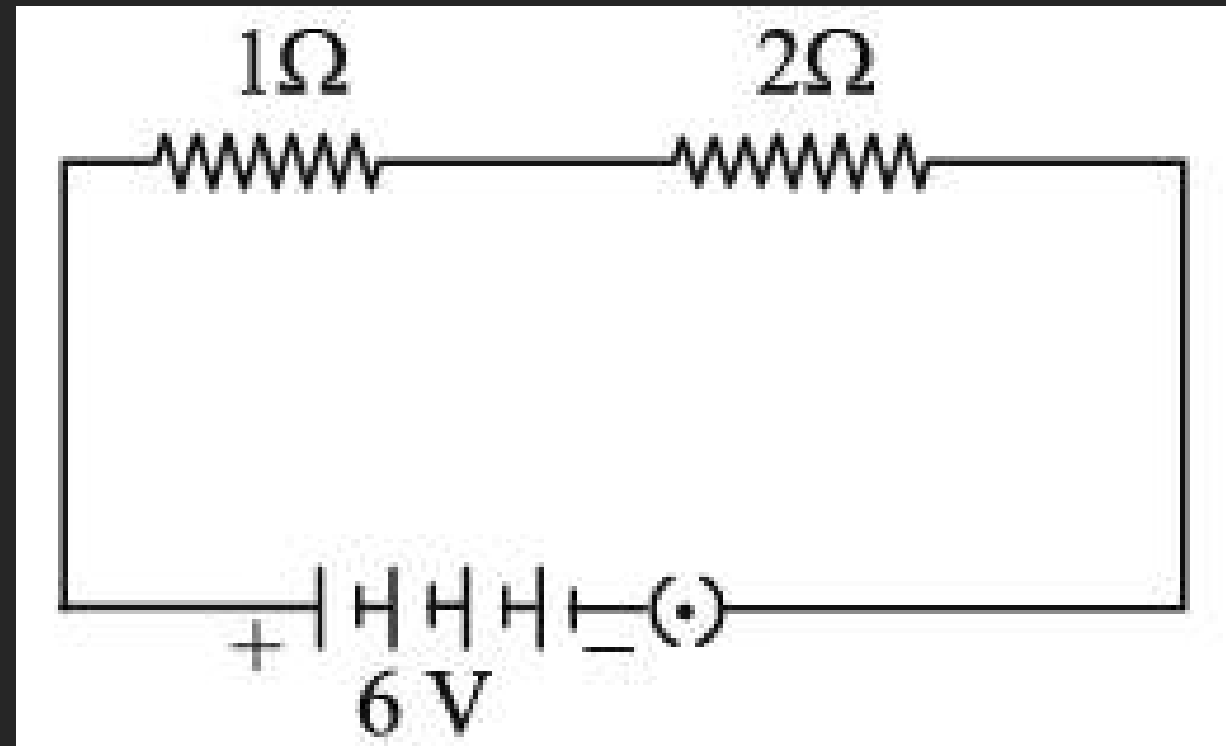
$$P = \frac{W}{t} = \frac{\Delta E}{t}$$

1 W = Electric power is said to be 1 watt when 1 ampere of current flows through a circuit under a potential difference of 1 volt.

- Practical units: 1 kilowatt (kW) = 1000 W

2024 (Board)

Q. In the given circuit calculate the power consumed in watts in the resistor of $2\ \Omega$:



2015, 2017, 2019, 2021 (Sample), 2023 (Board), 2024 (Board)

Q. An electric bulb is rated 220 V and 100 W. When it is operated on 110 V, the power consumed will be:

- (a) 220 W**
- (b) 75 W**
- (c) 50 W**
- (d) 25 W**

2017, 2019, 2021, 2023 (Sample), 2024 (Board)

Q. Two bulbs rated 60 W–220 V and 100 W–220 V are connected in parallel to 220 V supply. Find total power consumed.

2020,2023

Q. Two LED bulbs of 10 W and 5 W are connected in series. If the current flowing through the 5 W bulb is 0.005 A, the current flowing through the 10 W bulb is:

- (a) 0.02 A**
- (b) 0.001 A**
- (c) 0.005 A**
- (d) 0.0025 A**

Q. Consider two electric bulbs A and B having ratings 100 W–220 V and 40 W–220 V. Find the ratio of their resistances.

2015, 2018, 2020 Boards; 2022 Sample

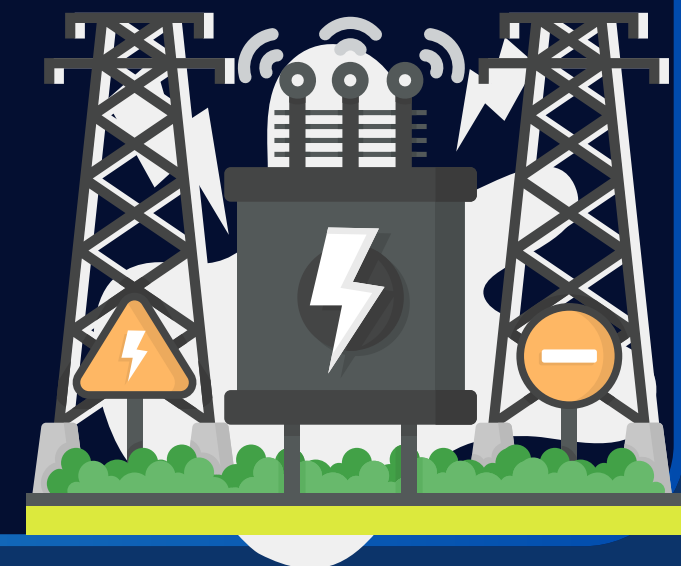
Q. An electric iron of power 1 kW is operated at 220 V. Which fuse wire of rating 1 A, 3 A, or 5 A must be used for it? Give a reason.

Electric Energy

Electric energy is the total work done or energy consumed by an electric device when a current flows through it for a given time.

Energy used by a circuit to allow current flow. It is the product of power and time, measured in **watt-hours (Wh)**.

$$E = P \times t = VIt = I^2Rt = \frac{V^2}{R} \times t$$



Commercial Unit of Energy:

One watt-hour is the energy used when 1 watt of power is consumed for 1 hour. The commercial unit of electric energy is the kilowatt-hour (kWh), also called a “unit.”

$$E = P \times t$$

$$\text{Joule} = \text{Watt} \times \text{Second}$$

$$J = Ws$$

$$E = \text{Kilowatt hour}$$

$$1\text{kWh} = 1000\text{Wh}$$

$$= 1000 \times 3600\text{Ws}$$

$$= 1\text{kWh} = 3.6 \times 10^6 \text{ J}$$



2016, 2018, 2020, 2022 (Sample), 2023 (Sample)

Q. In a house, 2 bulbs of 50 W each are used for 6 hours daily and an electric geyser of 1 kW is used for 1 hour daily. Calculate the total energy consumed in a month of 30 days and its cost at the rate of ₹ 8.00 per kWh.

2012, 2015 (Sample), 2017, 2019, 2020 (Sample), 2022, 2024 (Sample)

Q. A household uses the following appliances:

- **1 fan of 100 W for 10 hours per day**
- **2 tube lights of 40 W each for 5 hours per day**
- **1 refrigerator of 150 W running 24 hours per day**

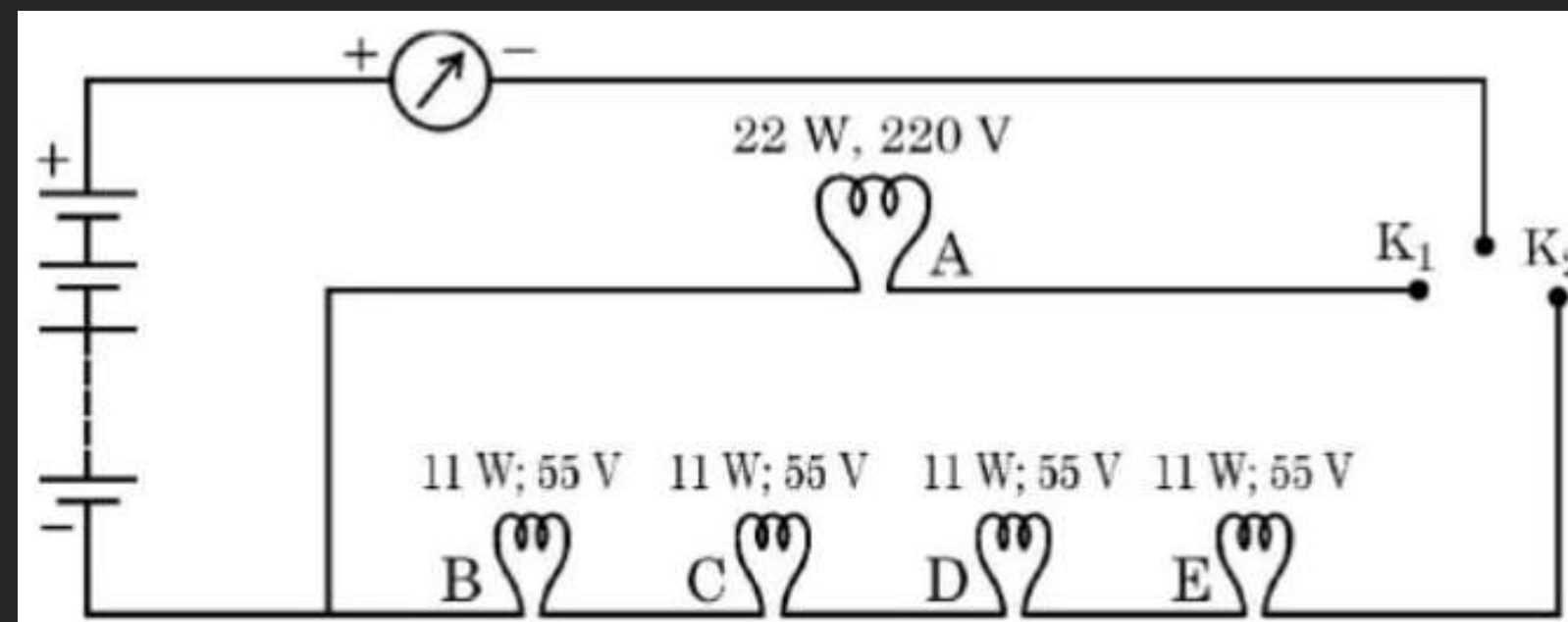
Find the total energy consumption in kWh per month (30 days) and the cost of energy at the rate of ₹5.00 per unit.

2024

**Q. An electric oven is designed to work on the mains voltage of 220 V. This oven consumes 11 units of electrical energy in 5 hours. Calculate :
(a) power rating of the oven (b) current drawn by the oven (c) resistance of the oven when it is red hot.**

Q. Case Based Question

In a domestic circuit, five LED bulbs are arranged as shown. The source voltage is 220 V, and the voltage and power rating of each bulb is given in the circuit diagram. Based on the diagram, answer the following:



- (a) State what happens when (i) key K_1 is closed, (ii) key K_2 is closed.
- (b) Find the current drawn by bulb B when it glows

Q. Case Based Question

(c) Calculate (i) the resistance of bulb B, and (ii) the total resistance of the combination of four bulbs B, C, D, and E.

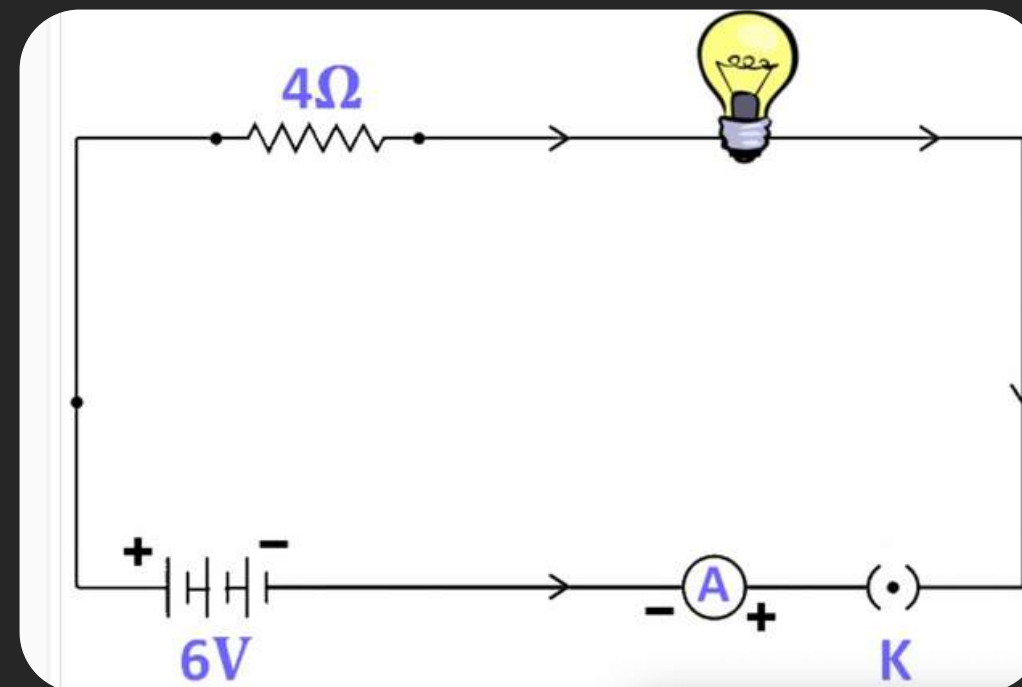
OR

(c) What would happen to the glow of all the bulbs in the circuit when keys K_1 and K_2 both are closed and the bulb C suddenly gets fused? Give a reason to justify your answer.

2011, 2013, 2015, 2017, 2018 (Sample), 2019, 2020 (Board), 2023 (Sample), 2024 (Board)

Q. An electric lamp of resistance $20\ \Omega$ and a conductor of resistance $4\ \Omega$ are connected to a $6\ \text{V}$ battery as shown in the circuit. Calculate:

- (a) the total resistance of the circuit**
- (b) the current through the circuit**
- (c) the potential difference across the (i) electric lamp and (ii) conductor**
- (d) the power of the lamp**



FORMULA SHEET

QUANTISATION OF CHARGE $Q = ne$

ELECTRIC CURRENT $I = \frac{Q}{T}$

POTENTIAL DIFFERENCE $V = \frac{W}{Q}$

OHM'S LAW $V \propto I$
 $V = IR$

RESISTANCE $R = \frac{V}{I}$

SERIES COMBINATION $R_s = R_1 + R_2 + R_3$

PARALLEL COMBINATION

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Joules Law of Heating $H = I^2 R t$

ELECTRICAL POWER $P = \frac{W}{t}$

Electric Energy $E = P \times t$



**“Even a small spark
can chase away
darkness – be that
spark in your own life.”**

